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ADNOC

ADNOC GROUP PROJECTS AND ENGINEERING

ARCHITECTURAL FINISHES AND FURNISHING

Specification

APPROVED BY:

10/02/2022

NAME: Abdulmunim Al Kindy
TITLE: Executive Director PT&CS
EFFECTIVE DATE:

AGES-SP-01-014

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GROUP PROJECTS & ENGINEERING / PT&CS DIRECTORATE

CUSTODIAN	Group Projects & Engineering / PT&CS
ADNOC	Specification applicable to ADNOC & ADNOC Group Companies

REVISION HISTORY

DATE	REV. NO	PREPARED BY (Designation / Initial)	REVIEWED BY (Designation / Initial)	ENDORSED BY (Designation / Initial)	ENDORSED BY (Designation / Initial)
21-Jan-2022	1	Kuriakose Molelputhenpura / SE-Civil-AGP Digitally signed by MANUR AJITH SHENOY DN: cn=MANUR AJITH SHENOY, o=ADNOC, ou=ADNOC GAS PROCESSING, email=mshenoy@adnoc.ae, c=AE Date: 2022.01.21 10:38:07 +04'00'	Mahmoud Abdel Hakim/ HOD Pipelines Eng. – GP&E Mahmoud Abdel Hakim Digitally signed by Mahmoud Abdel Hakim Date: 2022.01.21 10:44:51 +04'00' Reuben Yagambaram/ Mgr. Offshore Portfolio - GP&E Reuben Yagam baram Digitally signed by Reuben Yagambaram Date: 2022.01.21 11:46:59 +04'00'	Najem Qambar/ VP Group Eng.- GP&E 26/01/2022 Ali Al Breiki/ VP Upstream Projects- GP&E 26/01/2022	Ebraheem AlRomaithi / SVP- GP&E 09/02/2022

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In addition, Group Projects & Engineering is responsible for communication and distribution of any changes to this Specification and its version control.

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INTER-RELATIONSHIPS AND STAKEHOLDERS

a) The following are inter-relationships for implementation of this Specification:

   i. ADNOC Upstream and ADNOC Downstream Industry, Marketing and Trading Directorate.

   ii. ADNOC Onshore, ADNOC Offshore, ADNOC Sour Gas, ADNOC Gas Processing. ADNOC LNG, ADNOC Refining, Fertil, Borouge, Al Dhafra Petroleum, Al Yasat.

b) The following are stakeholders for the purpose of this Specification:

   i. ADNOC PT&CS Directorate

c) This Specification has been approved by the ADNOC PT&CS is to be implemented by each ADNOC Group company included above subject to and in accordance with their Delegation of Authority and other governance-related processes in order to ensure compliance.

d) Each ADNOC Group company must establish / nominate a Technical Authority responsible for compliance with this Specification.

DEFINITIONS

"**ADNOC**" means Abu Dhabi National Oil Company.

"**ADNOC Group**" means ADNOC together with each company in which ADNOC, directly or indirectly, controls fifty percent (50%) or more of the share capital.

"**Approving Authority**" means the decision-making body or employee with the required authority to approve Policies & Procedures or any changes to it.

"**Business Line Directorates**" or "**BLD**" means a directorate of ADNOC which is responsible for one or more Group Companies reporting to, or operating within the same line of business as, such directorate.

"**Business Support Directorates and Functions**" or "**Non- BLD**" means all the ADNOC functions and the remaining directorates, which are not ADNOC Business Line Directorates.

"**CEO**" means chief executive officer.

"**Group Company**" means any company within the ADNOC Group other than ADNOC.

"Specification" means this Architectural Finishes and Furnishing specification.

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1 GENERAL

1.1 Introduction

The purpose of this document is to describe the technical requirements for the furnishing and installation of plastering, dry walling, floor & wall tiling, skirting, cladding / lining panels, acoustical ceiling tile, resilient flooring, carpet, painting, wall coverings and accessories in buildings. It also defines the technical requirements for the furnishing and installation of all laboratory furniture, shop furniture, movable wall / partition systems and miscellaneous furniture, furnishings and equipment.

1.2 Purpose

This Specification provides minimum requirements for the architectural finishes and furnishings noted above and as outlined in the drawings and contract documents.

1.3 Definitions and Abbreviations

1.3.1 Definitions

The following defined terms are used throughout this Specification:

'**[PSR]**' indicates a mandatory Process Safety Requirement

'**COMPANY**' means ADNOC, ADNOC Group or an ADNOC Group Company, and includes any agent or consultant authorized to act for, and on behalf of the COMPANY.

'**CONTRACTOR**' means the parties that carry out all or part of the design, engineering, procurement, construction, commissioning, or management for ADNOC Projects. CONTRACTOR includes its approved MANUFACTURER(s), SUPPLIER(s), SUB-SUPPLIER(s) SUB-CONTRACTOR(s) and CONSULTANTS.

'**MANUFACTURER**' means the Original Equipment Manufacturer (OEM) or MANUFACTURER of one or more of the components(s) which make up a sub-assembly or item of equipment assembled by the main SUPPLIER or his nominated SUB-SUPPLIER.

'may' means a permitted option

'shall' indicates mandatory requirements

'should' means a recommendation

'**SUB-CONTRACTOR**' means any party engaged by the CONTRACTOR to undertake any assigned work on their behalf. COMPANY maintains the right to review all proposed SUB-CONTRACTORs; this right does not relieve the CONTRACTOR of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and the SUB-CONTRACTOR.

'SUPPLIER' means the party entering into a Contract with COMPANY to provide the materials, equipment, supporting technical documents and / or drawings, guarantees, warranties and / or agreed services in accordance with the requirements of the purchase order and relevant specification(s). The term SUPPLIER includes any legally appointed successors and / or nominated representatives of the SUPPLIER.

'**SUB-SUPPLIER**' means the sub-contracted SUPPLIER of equipment sub-components software and / or support services relating to the equipment / package, or part thereof, to be provided by the SUPPLIER. COMPANY maintains the right to review all proposed SUB-SUPPLIERS, but this right does not relieve

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the SUPPLIER of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and any individual SUB-SUPPLIER.

1.3.2 Abbreviations

Abbreviations used in this specification are shown in Table 1.1.

Table 1.1 List of Abbreviations

Abbreviations	
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
AVL	Approved Vendor List (COMPANY)
BS EN	British Standard, European Norm
BSI	British Standards Institution
HVAC	Heating, Ventilation, Air Conditioning
ISO	International Organization for Standardization
OC	On center
OSHA	Occupational Safety and Health Administration
QA / QC	Quality Assurance / Quality Control
UL	Underwriter's Laboratories

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SECTION A – GENERAL

2 REFERENCE DOCUMENTS

2.1 International Codes and Standards

The following Codes and Standards shall form a part of this specification. When an edition date is not indicated for a Code or Standard, the latest edition in force at the time of the contract award shall apply.

Applicability of UAE Codes may depend upon project location such as outside ADNOC concession areas and shall be agreed with COMPANY

UNITED ARAB EMIRATES CODES

New Abu Dhabi Building Codes

UAE Fire and Life Safety Code of Practice, General Headquarters of Civil Defence, Ministry of Interior, UAE

UNITED STATES CODES

International Building Code (IBC), except all sections and paragraphs related to accessibility

International Fire code (IFC)

International Residential Code (IRC)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A108	American National Standards Specifications for the Installation of Ceramic Tile
ANSI A118	American National Standards Specifications for mortar and grout
ANSI A108.4	Ceramic Tile Installed with Organic or Epoxy Adhesives
ANSI A108.5	Ceramic Tile Installed with Dry Set Portland Cement Mortar
ANSI A108.6	Ceramic Tile Installed with Chemical Resistant, Water Cleanable Tile-Setting and Grouting Epoxy
ANSI A136.1	Organic Adhesives for Installation of Ceramic Tile
ANSI A137.1	American National Standard Specifications for Ceramic Tile

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FEDERAL SPECIFICATIONS (FED. SPEC)

Fed. Spec. A-A-1922A	Shield, Expansion (Caulking Anchors, Single Lead)
Fed. Spec. A-A-1925A	Shield, Expansion (Nail Anchors)
Fed. Spec. A-A-55614	Shield, Expansion (Non-Drilling Expansion Anchors)
Fed. Spec. CCC-W-408D	Wall covering, Vinyl-coated
Fed. Spec. CFFA-W-101D	Vinyl coated fabric wallcovering – Purpose
Fed. Spec. DDD-C-0095	Carpet and Rugs, Wool, Nylon, Acrylic, Modacrylic, Polyester, Polypropylene
Fed. Spec. SS-W-40A	Wall Base: Rubber, and Vinyl Plastic
Fed. Spec. UU-B-790A	Building Paper, Vegetable Fiber (Kraft, Waterproofed, Water Repellent and Fire Resistant)
Handbook	CISCA Ceiling Systems Handbook

US DEPT. OF COMMERCE, BUREAU OF STANDARDS, PRODUCT STANDARDS (PS)

FED PS-20	American Softwood Lumber Standard
APA PS-1	Structural Plywood

TILE COUNCIL OF NORTH AMERICA (TCNA)

TCNA EJ171	Expansion Joints, Vertical & Horizontal
TCNA F113	Concrete Subfloor, Interior
TCNA F132	Concrete Subfloor, Interior (Chemical Resistant)
TCNA W202	Masonry Walls, Exterior & Interior
TCNA W211	Masonry or Concrete Walls, Interior
TCNA W212 / W213	Handbook for Ceramic Tile Installation
TCNA W223	Walls with Solid Backing, Interior
TCNA W244	Cementitious board units over wood or metal stud
TCNA Handbook	Handbook for Ceramic, Glass, and Stone Tile Installation

NATIONAL WOODWORK MANUFACTURERS ASSOCIATION (NWMA)

FLAT GLASS MARKETING ASSOCIATION (FGMA)

CONSUMER PRODUCT SAFETY COMMISSION (CPSC)

CPSC 16 / CFR 1201 Consumer Product Safety Commission Safety Standard for Architectural Glazing Materials

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

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NFPA 80	Standard for Fire Doors and Other Opening Protectives
NFPA 253	Standard Method of Test for Critical Radiant Flux of Floor Covering Systems Using a Radiant Heat Energy Source
NFPA 5000	Building Construction and Safety Code

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA) STANDARD

LD-3	High-Pressure Decorative Laminates

NATIONAL PARTICLE BOARD ASSOCIATION STANDARDS

ANSI A208.1	Particleboard
ANSI A208.2	Medium Density Fiberboard (MDF) for Interior Applications

NATIONAL WOODWORK MANUFACTURER'S ASSOCIATION (NWMA) STANDARD

NWMA 1.S.1	Series for Wood Flush Doors

ARCHITECTURAL WOODWORK INSTITUTE (AWI) PUBLICATIONS

AWI	Quality Standards

UNITED STATES GYPSUM (USG)

USG	Gypsum Construction Handbook

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC SP-2	Steel Structures Painting Manual: Hand Tool Cleaning

STEEL DOOR INSTITUTE (SDI) STANDARD

ANSI / SDI A250.8	Specifications for Standard Steel Doors and Frames (SDI-100)

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA) STANDARD

AAMA 501	Methods of Test for Exterior Walls

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC 6	Test Method for Colorfastness to Acids and Alkalis.
AATCC 16.1, 16.2 & 16.3	Test Method for Colorfastness to Light
AATC 122	Test Method for Carpet Soiling: Service Soiling
AATCC 129	Test Method for Colorfastness to Ozone in the Atmosphere Under High Humidities
AATCC 134	Test Method – Electrostatic Propensity of Carpets
AATCC 164	Test Method – Colorfastness to Oxides of Nitrogen in the Atmosphere Under High Humidities
AATCC 165	Test Method – Colorfastness to Crocking: Textile Floor Coverings – Crockmeter
AATCC 174	Test Method – Anti-microbial Activity Assessment of new carpets
AATCC 175	Test Method – Stain Resistance: Pile Floor Coverings

BUILDER'S HARDWARE MANUFACTURERS ASSOCIATION (BHMA) STANDARDS

ANSI / BHMA A156.1	Butts and Hinges
ANSI / BHMA A156.2	Bored and Preassembled Locks and Latches
ANSI / BHMA A156.3	Exit Devices
ANSI / BHMA A156.4	Door Controls – Closers
ANSI / BHMA A156.9	Cabinet Hardware
ANSI / BHMA 156.16	Auxiliary Hardware
ANSI / BHMA 156.21	American National Standard for Thresholds

CHEMICAL FABRICS AND FILM ASSOCIATION (CFFA)

CFFA W101D	Quality Standards for Vinyl Coated Fabric Wall coverings

CEILING AND INTERIOR SYSTEMS CONSTRUCTION ASSOCIATION (CISCA)

Recommended Test Procedures for Access Floors

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AMERICAN HARDBOARD ASSOCIATION (AHA) STANDARD

AHA A135.4	Basic Hardboard

BRITISH STANDARDS INSTITUTION (BS)

BS 1199	Specifications for building sands from natural sources
BS 1336	Specification for knotting
BS 5325	Installation of textile floor coverings. Code of Practice
BS 5974	Planning, design, setting up and use of temporary suspended access equipment - code of practice
BS 6150	Painting of Buildings - Code of Practice
BS 6319	Testing of Resin Composition for Use in Construction - Methods for measurement of modulus of elasticity in flexure and flexural strength
BS 8000	Workmanship on building sites. Code of practice for decorative wall coverings and painting.
BS 8203	Code of Practice for Installation of Resilient Floor Coverings
BS 8298	Code of practice for design and installation of natural stone cladding and lining

EURO CODES (BS EN)

BS EN 197	Part 1: Cement: Composition, specifications and conformity criteria for common cements
BS EN 459	Building lime - Definitions, specifications and conformity criteria
BS EN 490	Concrete roofing tiles and fittings for roof covering and wall cladding – Product Specifications
BS EN 491	Concrete roofing tiles and fittings for roof covering and wall cladding – Test Methods
BS EN 1008	Mixing Water for Concrete. Specification for sampling, testing and assessing the suitability of water, including water recovered from processes in the concrete industry, as mixing water for concrete.
BS EN 10143	Continuously hot-dip coated steel sheet and strip. Tolerances on dimensions and shape
BS EN 10346	Continuously Hot-Dip Coated Steel flat products – Technical delivery conditions
BS EN 12004	Adhesives for ceramic tiles - Requirements, assessment and verification of constancy of performance, classification and marking

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BS EN 12811	Scaffolds — Performance requirements and general design
BS EN 13139	Aggregates for Mortar
BS EN 13150	Workbenches for laboratories in educational institutions. Dimensions, safety and durability requirements and test methods
BS EN 13501	Fire classification of construction products and building elements - Classification using data from reaction to fire tests
BS EN 13658	Metal Lath and Beads, Definitions, Requirements and Test Methods
BS EN 13888	Grout for tiles Requirements, evaluation of conformity, classification and designation
BS EN 13964	Suspended ceilings - requirements and test methods
BS EN 14056	Laboratory furniture. Recommendations for design and installation
BS EN 14259	Adhesives for floor covering. Requirements for mechanical and electrical performance
BS EN 17668	Adhesives for floor coverings. Preparation of adhesive application. Test method for the determination of excessive humidity in subfloors
BS EN 14411	Ceramic tiles. Definitions, classification, characteristics, assessment and verification of constancy of performance and marking
BS ISO 3864	Graphical Symbols - Safety Colours and Safety Signs
BS EN ISO 4599	Plastics. Determination of resistance to environmental stress cracking (ESC). Bent strip method
BS EN ISO 9001	Quality Management Systems - Requirements
BS EN ISO 9004	Quality Management – Quality of an organization - Guidance to achieve sustained success
BS EN ISO 10545	Ceramic Tiles
BS EN ISO 10581	Resilient floor coverings. Homogeneous poly(vinyl chloride) floor covering. Specifications
BS EN ISO 22631	Adhesives. Test method for adhesives for floor and wall coverings. Peel test
BS EN ISO 22632	Adhesives. Test method for adhesives for floor and wall coverings. Shear test
BS EN ISO 22633	Adhesives. Test methods for adhesives for floor coverings and wall coverings. Determination of dimensional changes of a linoleum floor covering in contact with an adhesive
BS EN ISO 22636	Adhesives. Adhesives for floor coverings. Requirements for mechanical and electrical performance

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INTERNATIONAL ELECTROTECHNICAL COMMISSION STANDARD

IEC 60364	Low-voltage electrical installations

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO))

ISO 717	Acoustics - Rating of Sound insulation in Buildings and of building elements – Part 1: Airborne sound insulation
ISO 19011	Guidelines for Quality and / or Environmental Management Systems Auditing
ISO 80000-8	Quantities and units – Part 8: Acoustics

TESTING AND LOSS PREVENTION

Underwriters Laboratories INC (UL)

UL10 B Standard for Fire Tests of Door Assemblies

Factory Mutual Engineering Corporation (FM Global)

OFFSHORE SAFETY AND RISK MANAGEMENT

DNVGL-OS-D301 Offshore Standards, Fire Protection

FTP Code 2010 International Code for Application of Fire Test Procedures

IMO SOLAS International Maritime Organization Safety of Life at Sea Conventions

NORSOK S-001 Technical safety

NORSOK-S-002 Working environment

ASTM STANDARDS (AMERICAN SOCIETY FOR TESTING AND MATERIALS)

ASTM A90 / A90M	Standard Test Method for Weight (Mass) of Coating on Iron and Steel Articles with Zinc or Zinc-Alloy Coatings
ASTM A641 / A641M	Standard Specification for Zinc-coated (Galvanized) Carbon Steel Wire
ASTM A653 / A653M	Standard Specification for Steel Sheet, Zinc Coated (Galvanized) or Zinc Iron Alloy Coated (Galvannealed) by the Hot Dip Process
ASTM A924 / A924M	Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process

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ASTM A1008 / A1008M	Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable
ASTM B221	Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM C5	Standard Specification for Quicklime for Structural Purposes
ASTM C35	Standard Specification for Inorganic Aggregates for Use in Gypsum Plaster
ASTM C144	Standard Specification for Aggregate for Masonry Mortar
ASTM C150 / C150M	Standard Specification for Portland Cement
ASTM C206	Standard Specification for Finishing Hydrated Lime
ASTM C207	Specification for Hydrated Lime for Masonry Purposes
ASTM C272 / C272M	Standard Test Method for Water Absorption of Core Materials for Sandwich Constructions
ASTM C413	Standard Test Method for Absorption of Chemical Resistant Mortars, Grouts, Monolithic Surfacing, and Polymer Concretes
ASTM C423	Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method
ASTM C475 / C475M	Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board
ASTM C503 / C503M	Standard Specification for Marble Dimension Stone
ASTM C612	Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM C615	Standard Specification for Granite Dimension Stone
ASTM C635 / C635M	Standard Specification for the Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-in Panel Ceilings
ASTM C645	Standard Specification for Nonstructural Steel Framing Members

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ASTM C650	Standard Test Method for Determination of Resistance to Chemical Substances
ASTM C722	Standard Specification for Chemical-Resistant Monolithic Floor Surfacings
ASTM C754	Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products
ASTM C840	Standard Specification for Application and Finishing of Gypsum Board
ASTM C841	Standard Specification for Installation of Interior Lathing and Furring
ASTM C847	Standard Specification for Metal Lath
ASTM C920	Standard Specification for Elastomeric Joint Sealants
ASTM C926	Standard Specification for Application of Portland Cement-Based Plaster
ASTM C1002	Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs.
ASTM C1047	Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base
ASTM C1063	Standard Specification for Installation of Lathing and Furring to Receive Interior and Exterior Portland Cement-Based Plaster
ASTM C1172	Standard Specification for Laminated Architectural Flat Glass
ASTM C1242	Standard Guide for Selection, Design, and Installation of Dimension Stone Attachment Systems
ASTM C1354	Standard Test Method for Strength of Individual Stone Anchorages in Dimension Stone
ASTM C1396 / 1396M	Standard Specification for Gypsum Board
ASTM C1670 / C1670M	Standard Specification for Adhered Manufactured Stone Masonry Veneer Units

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ASTM D16	Standard Terminology for Paint, Related Coatings, Materials, and Applications
ASTM D41 / D41M	Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing
ASTM D312 / D312M	Standard Specification for Asphalt Used in Roofing
ASTM D1621	Standard Test Method for Compressive Properties of Rigid Cellular Plastics
ASTM D1623	Standard Test Method for Tensile and Tensile Adhesion Properties of Rigid Cellular Plastics
ASTM D2485	Standard Test Methods for Evaluating Coatings for High Temperature Service
ASTM D2859	Standard Test Method for Ignition Characteristics of Finished Textile Floor Covering Materials
ASTM D3574	Tests involving flexible cellular foam materials.
ASTM D3730	Standard Guide for Testing High-Performance Interior Architectural Wall Coatings
ASTM D4258	Standard Practice for Surface Cleaning Concrete for Coating
ASTM D4259	Standard Practice for Preparation of Concrete by Abrasion Prior to Coating Application
ASTM D4260	Standard Practice for Liquid and Gelled Acid Etching of Concrete
ASTM D4261	Standard Practice for Surface Cleaning Concrete Masonry Units for Coating
ASTM D4801	Standard Specification for Polyethylene Sheeting in Thickness of 0.25 mm (0.010 in.) and Greater
ASTM D4946	Standard Test Method for Blocking Resistance of Architectural Paints
ASTM D5146	Standard Guide to Testing Solvent-Borne Architectural Coatings
ASTM D5324	Standard Guide for Testing Water-Borne Architectural Coatings

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ASTM D6577	Standard Guide for Testing Industrial Protective Coatings
ASTM E72	Standard Test Methods of Conducting Strength Tests of Panels for Building Construction
ASTM E84	Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E90	Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements
ASTM E119	Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E648	Standard Test Method for Critical Radiant Flux of Floor Covering Systems Using a Radiant Heat Energy Source
ASTM E662	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials
ASTM E1264	Standard Classification for Acoustical Ceiling Products
ASTM E1414 / E1414M	Standard Test Method for Airborne Sound Attenuation Between Rooms Sharing a Common Ceiling Plenum
ASTM F150	Standard Test Method for Electrical Resistance of Conductive and Static Dissipative Resilient Flooring
ASTM F710	Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring
ASTM F793 / F793M	Standard Classification of Wall Covering by Use Characteristics
ASTM F1066	Standard Specification for Vinyl Composition Floor Tile
ASTM F1344	Standard Specification for Rubber Floor Tile
ASTM F1860	Standard Specification for Rubber Sheet Floor Covering with Backing
ASTM F1861	Standard Specification for Resilient Wall Base
ASTM F2169	Standard Specification for Resilient Stair Treads

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2.2 ADNOC Specifications

The following ADNOC general specifications, as available with respective Group Company and as applicable to the Project, shall form a part of this specification. When an edition date is not indicated for a document, the latest edition in force at the time of contract award shall apply.

Concrete Supply and Construction Specification	AGES-SP-01-001
Structural Steel Supply, Fabrication and Erection Specification	AGES-SP-01-002
Architectural Design Basis	AGES-SP-01-013
Masonry Wall and Thermal & Moisture Protection	AGES-SP-01-017
Marine and Coastal Structures: General Requirements	AGES-SP-01-023
Building Safety Specification	AGES-SP-03-003
Human Factors Engineering Specification	AGES-SP-03-004
Painting & Coating Specification	AGES-SP-07-004
HVAC Design Basis	AGES-SP-14-001
HVAC Equipment	AGES-SP-14-002
HVAC Ductwork, Pipework and Accessories	AGES-SP-14-003
Basic Engineering Design Data (BEDD)	AGES-GL-16-003
Ergonomics Hazards	HSE-OH-ST11
Green Buildings Guidelines	AGES-GL-01-003
Architectural Specialties, Metal, Wood and Plastics	TBC
Doors and Windows	TBC

2.3 Standard Drawings

The following ADNOC standard drawings, as available with respective Group Company and as applicable to the Project, shall form a part of this specification. When an edition date is not indicated for a document, the latest edition in force at the time of contract award shall apply.

Building Architectural works - General Notes, Abbreviations & Symbols	TBC
Typical Architectural Details	TBC

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Typical Detail of Concrete Block walls	TBC

2.4 Other References

Not applicable.

3 DOCUMENT PRECEDENCE

The specifications and codes referred to in this shall, unless stated otherwise, be the latest approved issue at the time of contract award.

It shall be the CONTRACTOR's responsibility to be, or to become, knowledgeable of the requirements of the referenced Codes and Standards.

The CONTRACTOR shall notify the COMPANY of any apparent conflict between this specification, the related data sheets, the Codes and Standards and any other specifications noted herein.

Resolution and / or interpretation precedence shall be obtained from the COMPANY in writing before proceeding with the design / manufacture.

a. UAE Statutory requirements

b. ADNOC HSE Standards

c. Equipment datasheets and drawings

d. Project Specifications and standard drawings

e. Company Specifications

f. National / International Standards

4 SPECIFICATION DEVIATION / CONCESSION CONTROL

Any technical deviations to the Purchase Order and its attachments including, but not limited to, the COMPANY's General Specifications shall be sought by the CONTRACTOR only through technical deviation request format. Technical deviation requests require COMPANY'S review / approval, prior to the proposed technical changes being implemented. Technical changes implemented prior to COMPANY approval are subject to rejection.

5 DESIGN CONSIDERATIONS / MINIMUM DESIGN REQUIREMENTS

Refer to Section B of this specification.

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SECTION B – TECHNICAL REQUIREMENTS

6 TECHNICAL REQUIREMENTS

6.1 Materials

a) Colors and finishing tints are generally not specified in this document. When necessary, manufacturer's standard palette shall be provided by CONTRACTOR for the choice of the COMPANY.

b) The CONTRACTOR shall submit an 'Affidavit of Compliance' before beginning construction. The Affidavit shall confirm that all material meets the requirements of this specification.

6.2 Plastering

6.2.1 General

6.2.1.1 Scope of Work

a) This section covers the plasterworks. The design and execution shall be state-of-the-art, functional and complete in all parts in accordance with this Specification.

b) This section shall apply to all parts of the work in which plaster is to be used. The work shall include all necessary labor, materials, equipment and auxiliary works as required. Plasterwork shall be done in three coats. The minimum thickness of finished plaster shall be:

i. With exterior & interior plastering 15-20 mm, to COMPANY approval

ii. With interior plastering 15 mm, in cases where walls are to be finished with wall tiles

iii. Ceiling plastering 12 mm, in cases where finished form is not acceptable

c) Plastering shall be coordinated with other building, electrical, ventilating and plumbing work. In general, the edges of all openings through the plaster shall be protected with metal grounds or casings made accurately.

6.2.1.2 Related Work Specified Elsewhere:

a) See AGES-SP-01-017 Masonry Wall and Thermal & Moisture Protection.

b) See Company Standard Drawing for typical detail of Concrete Block walls.

c) See Painting, Section 6.8 herein.

d) See APPENDIX A - Typical Finishing Schedules, AGES-SP-01-013 Architectural Design Basis.

6.2.1.3 Submittals

a) Complete specifications, product data, samples with color & pattern and Manufacturer's data sheets on each product to be used under this section shall be submitted to the COMPANY for review and approval, including the following:

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b) Submit the shop drawing showing locations of all expansion and control joints in elevation and details of such joints along with type of metal accessories used in the work.

c) Submit full size mock-up panel over masonry substrate. Approved sample shall be available at the job site with final sign off by the COMPANY.

6.2.1.4 Delivery and Storage

a) All materials shall be delivered in the original packages bearing the name of the manufacturer and the brand.

b) Plaster and other cementations materials shall be kept dry until ready to be used; they shall be stored in an enclosed shelter, kept off the ground, and away from sweating walls and other damp surfaces. Metal goods shall be protected against rusting. Damaged or deteriorated materials shall be removed from the site of the work.

6.2.1.5 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

6.2.2 Materials

Metal Corner Beads	Expansion type fabricated from 0.50 mm (28 gauge) galvanized steel
Casing beads and other metal accessories	0.60 mm (26 gauges) galvanized steel perforated or expanded flanges or clips
Quicklime	ASTM C5, pulverized
Hydrated lime	ASTM C206, Type S / BS EN 459-1
Lime Putty	Quicklime, thoroughly slaked and stored for 1 week, or hydrated lime saturated and stored for 24 hours; kept moist until used
Portland cement	Standard brand of Portland Cement conforming to ASTM C150, Type 1, (BS EN 197-1) white for finish coat
Sand	To BS EN 13139, graded 3 mm down, washed free of chlorides and soluble organic matter. Tested to BS EN 1008

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Water	To be clean, potable and free from sulphides, chlorides and soluble organic matter. Tested to BS EN 1008. Water shall be 'mix quality' as defined in Company Specification AGES-SP-01-001
Expanded Metal Lath	ASTM C847, zinc coated (galvanized) except as modified by ASTM C841 and this specification. Self-furring where applied over solid backing. Type of Expanded Metal Lath shall be approved by COMPANY
Stucco Mesh	Flat expanded diamond mesh pattern, with openings approximately 38 by 75 mm with backing
Admixtures and additives	Cement containing additives may only be used after COMPANY approval
Bonding agent	Bonding agent shall be compatible with background and finish, designated to be suitable for internal or external use may only be used after COMPANY approval
Stone Chip Render	Stone chippings for exposed aggregate rendering shall be granite, white stone or marble chippings, graded from 3 to 5 mm, may only be used after COMPANY approval
Pre-mixed Plaster	Cement based plaster contains mineral filler material as aggregate and Portland cement as main binder, with additives for improved compressive strength and reduced shrinkage
General	Provide Number 1 white silica with 100 percent passing a Number 20 sieve, conforming to ASTM C35 (BS 1199, table 1) for finish coat. Sand shall be clean, free from organic material, with no more than 5 percent clay, silt, or other impurities and shall contain no salt or alkali

6.2.3 Execution

6.2.3.1 Installation of Metal Lath and Accessories / Metal Plastering bases (Lath) Locations

a) Where plaster is required on concrete or masonry bases, metal plastering lath is required and shall be shown on the drawings. The installation shall be completed in accordance with ASTM C841 as follows:

i. On exterior wall framing:

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Where metal plastering bases are used as a base for exterior cement plaster over concrete or masonry bases, use wire lath or stucco mesh with water resistant backing joints and also at a point of contact of two different materials.

ii. On interior wall framing:

Use flat diamond mesh lath and use lath with water resistant backing in wet areas.

6.2.3.2 Applying Metal Plastering Lath or Bases

a) Form true surfaces, straight or in fair curves where shown, without sags or buckles and with long dimension of lath at right angles to direction of supports.

b) Lath with backing shall be applied to produce a paper to paper and metal to metal lap at ends and sides of adjacent sheets, whether full sheets or less than full sheets are used.

c) Backing shall be lapped 50 mm for both horizontal and vertical laps or as shown on the shop drawings.

d) Horizontal laps shall be formed ship lap fashion to conduct water to the outside, and over flashing or waterproofing.

e) Metal plastering bases shall not be continuous through expansion and control joints but shall be stopped at each side.

f) Attach metal lath directly to masonry and concrete with hardened nails, power actuated drive pins or other approved fasteners. Fasteners shall be located at the dimples or crimps only at every 300 mm canters.

6.2.3.3 Installing Plastering Accessories

Accessories:

Set plastering accessories plumb, level and true to line, neatly mitred at corners and intersections, and securely attach to supporting surfaces as specified for metal lath.

Corner Beads:

a) Install at all vertical and horizontal external plaster corners, as required to establish grounds, and as shown on shop drawings.

b) Install at interior corners of walls, partitions, and other vertical surfaces to be plastered, except where metal lath is carried around angles.

c) Fasten only as necessary to retain position during plastering.

d) Omit cornerites or other reinforcing angles at junction of new plastered walls with existing plastered walls at locations where casing beads are specified.

Strip Lath:

a) Install metal lath strips centred over joints between dissimilar materials, such as concrete masonry units, concrete, and joints with metal lath on framing or furring, where both such surfaces are required to be plastered and are in contact with each other in same plane, except where expansion joints and casing beads are required.

b) Fasten strip lath over the base as shown on shop drawings.

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Casing Beads:

a) Install casing beads where shown and at following locations where plaster terminates to provide finish trim.

b) Where plaster terminates against no plastered surfaces such, concrete, wood, etc.

c) Around perimeter of openings except where edge is covered by flanges.

d) Where plaster for new walls or furring (vertical or horizontal) terminates against existing construction.

e) Both sides of expansion and control joints or as shown on shop drawings.

Control Joints:

a) Where control joints are placed parallel to framing members, install joints within 100 mm of the framing member or as shown on drawings.

b) Install control joints only to the edges of abutting sheets of lath so that the lath is not continuous or tied across the joint.

c) Joints shall extend the full width and height of the wall or length of soffit / ceiling plaster membrane.

6.2.3.4 Installation of Plaster

a) Provide Portland cement-based plaster where cement plaster is shown and three coat works shall be used over all metal plastering bases, with or without solid backing. The minimum thickness of plaster is measured from face of lath, masonry, or concrete to finished plaster. Metal grounds shall be used to regulate the thickness.

b) Proportioning of various plaster mixes, plaster thickness and plaster application shall be in accordance with ASTM C926, except as otherwise specified.

   i. Use air entrained plaster for all exterior work

   ii. Use coloring pigments for finish coat when integral color other than white is specified

   iii. Factory prepared finish coat: add water, mix, and apply as specified by manufacturer

   iv. Color of finish coat shall be natural cement color when painted or other coating is specified, and finish coat shall be smooth trowelled / sand float / machine dash / textured as selected by COMPANY

c) Mechanically mixed plaster materials shall be used at the project site; do not hand mix except where small amounts are needed, using less than one bag of plaster material and prior approval shall be taken from COMPANY for the same.

d) Premixed plaster and clean water shall be mixed on site with portable electrical mixer or other equipment according to manufacturer's recommendations. Accurate amount of water shall be used to ensure effective mixing and consistent quality.

e) Where specified, slurry Coat shall be 400 kgs of cement to one cubic meter of sand. Second and third coats shall be 300 kgs of cement and 50 kgs of lime to one cubic meter of sand.

6.2.3.5 Preparation

a) CONTRACTOR shall carefully inspect to ensure that the following conditions do not exist, before plastering:

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i. Surface unevenness

ii. Efflorescence

iii. Stress and settlement cracks over smooth surfaces

iv. Oily surfaces (e.g. through form oil)

b) Masonry surfaces shall be prepared and cleaned by removing loose particles, dust adhering to mortar, grease, oil, efflorescence and other foreign matters, and shall be sprayed with water before plaster is applied. Use bonding agent. Concrete surfaces shall be carefully roughened and cleaned with water under pressure.

c) Ensure that the block wall joints have been keyed and that the walls are acceptable to start the plastering work. All work shall be in compliance with Masonry Wall and Thermal & Moisture Protection AGES-SP-01-017.

d) Install metal lath as required with stainless steel fasteners and lapped joints tied with galvanized wire. Cover with self-furring lath where required to keep the total plaster thickness required. Lath, if used, shall be flat against the wall with no bulges and shall be uniformly secured.

e) Metal grounds and other accessories, such as corner beads and metal casings, shall be carefully examined to see that they are straight, curved, plumb, level, square or true to the required angles as necessary in each case, before plaster is applied. Masonry surfaces shall be checked for excessive suction and if necessary, dampened before plastering. Install corner (angle) beads at all openings and fasten with stainless steel fasteners. Install plaster grounds (stops) to finish the plaster as required and fasten with stainless steel fasteners.

f) Prior to commencing plastering, check all that overhead work for all trades is complete and that the work area is free from excessive moisture. Following the installation of all light fittings and diffusers, check ceiling and make good any damage.

g) Plastering shall not commence until all mechanical and electrical services, conduits, pipes and fixtures have been installed.

6.2.3.6 Application

a) The first scratch coat or dash-bond coat: a thick wet mixture of Portland cement and water, with or without aggregate, dashed onto the surface of a plaster base of concrete or concrete block surfaces to improve the mechanical key for subsequent plaster coats. Allow at least one day before succeeding coat is applied. The thickness of the scratch coat shall be not less than 10 mm. The scratch coat shall set firm and hard before the brown coat is applied. The thickness of the brown coat shall be not less than 7 mm.

b) The brown coat shall be brought out to grounds, straightened to a true surface with rod and darby, and left rough, ready to receive the finish coat. The brown coat shall set dry and hard before finish coat is applied.

c) Scratch and brown coats for Portland cement plaster shall each be moist cured with water for at least 24 hours or as per manufacturer's instruction.

d) Finish Coat: The finish coat shall be laid out to permit the completion of an entire surface in one operation. Finish coat shall be applied to minimum thickness of 3 mm, or in such thickness as may be necessary to ensure the full thickness achieved as specified and shall have a sand float finish free from slick spots, blemishes, and irregularities. Brown coat shall be dampened evenly to provide suction before applying finish coat.

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e) Finishes shall be applied so that all surfaces are straight, true, smooth, even, plumb and flush with grounds. The use of excessive water shall be avoided.

f) Mechanically applied plaster: where plaster will be applied mechanically the first coat shall be 0.7 - 1 cm thick and shall avoid movement of metal accessories when the pressure mortar coat is applied. Immediately after setting, a second coat of approximately 1 cm thickness shall be applied by hand, smoothing any irregularities and finally treated.

g) Cement based premixed plaster shall be applied with trowel or spraying machine in several coats. Application method, thickness, number of coats and drying time between coats shall be strictly in accordance with manufacturer's recommendations.

h) Internal walls specified to be plastered shall be plastered up to the slab soffit level including the part above false ceiling (if any).

6.2.3.7 Protection

a) All plaster shall be protected from premature or uneven drying until thoroughly set. Portland cement plaster shall be moist cured for at least 48 hours.

b) The CONTRACTOR shall use drop cloths and other protective measures to protect all surfaces from accidental spraying, spattering, or spilling of plaster. Any damage resulting from his operation shall be repaired. All plaster deposited on surfaces which are not to be plastered shall be removed as soon as possible to the satisfaction of the COMPANY.

6.2.3.8 Patching

a) Plaster showing over-sanding, cracks, blisters, pits, checks, or discoloration will not be acceptable. Such plaster shall be removed and replaced with new plaster. Patching of defective work will be permitted only when acceptable to the COMPANY, and such patching shall match adjacent work in texture and color.

b) The CONTRACTOR shall point up and patch as necessary around grilles and other openings, leaving all work in a neat and finished condition.

6.2.3.9 Clean-up

At the completion of the finish plastering work, all plaster daubs shall be cleaned from beads, screens, and metal trim. All rubbish shall be removed from the building and floors shall be left broom clean. Excess material, scaffolding, tools, and other equipment shall be removed from the building and the project site.

6.3 Gypsum Drywall

6.3.1 General

6.3.1.1 Scope of Work

a) This section covers the minimum requirements for the method of installation and finishing of gypsum board partition walls, including related items and accessories. The work shall include all labor, materials, supervision and equipment for the installation and finish of all Drywall as per drawings.

b) All work shall be in accordance with ASTM C754 and ASTM C840

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c) Drywall construction shall be coordinated with other building, electrical, heating, ventilating, and plumbing work. In general, all corners and all exposed edges and openings through wallboard shall be protected with metal casings made accurately to size and dimension.

6.3.1.2 Related Work Specified Elsewhere:

a) See Company Specification for Architectural Specialties, Metal, Wood and Plastics

b) See Painting, Section 6.8 herein

c) See Wall covering, Section 6.9 herein

d) Architectural Design Basis AGES-SP01-013, see 'APPENDIX A - Typical Finishes Schedule'

e) See Company Standard drawing for Typical Architectural Details

6.3.1.3 Submittals

The following shall be submitted to the COMPANY for review and approval:

a) Submit manufacturer's technical data and drawings illustrating the details of the system and the manufacturer's recommended installation instructions. System and installation shall be in accordance with the requirements of Architectural Design Basis AGES-SP-01-013.

b) Manufacturer's certification of product compliance with codes and standards.

c) Gypsum Board Finish Mock-ups: Before finishing gypsum board assemblies, install mock-up sample of at least 100 square feet (9.3 sq. m) in surface area to demonstrate aesthetic effects and qualities of materials and execution.

d) VOC Classification: CONTRACTOR to submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013.

6.3.1.4 Delivery and Storage

a) Materials shall be delivered in original packages bearing the name of the manufacturer and the brand.

b) Store materials, keep dry, and protect against damage from weather, direct sunlight, surface contamination, corrosion, construction traffic and other causes.

c) Stack gypsum panels on level surface to prevent sagging.

d) Adhesives shall be protected from freezing and metal goods protected from rusting.

6.3.1.5 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

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6.3.2 Materials

Metal Studs	Typically ASTM A653 / A653M; galvanized steel, 0.9 mm (20 gauge), 90 mm deep channel type screw studs, with top and bottom tracks. Metal Studs: ASTM C645, ST22 100 mm (4 inch), 1 mm (20 gauge). Specification to be agreed with COMPANY per application
Furring Channels	Typically ASTM A653 / A653M; galvanized steel, 20 mm deep, 35 mm face width, screw furring channels with extended nailing flanges. Furring channels. 40 mm galvanized cold rolled steel channel section 0.7 mm thick. Specification to be agreed with COMPANY per application
Metal Runners	Same thickness as the channels or metal studs
Corner Beads	Type CB; galvanized steel, 30 mm by 30 mm perforated flanges
Casing Beads	Type LS; galvanized steel, with perforated flanges
Control Joint	Zinc or galvanized steel, with 20 mm wide V-shaped opening and perforated flanges
Joint Compound	ASTM C475 / C475M premixed paste, to manufacturer's recommendation
Joint Tape	ASTM C475 / C475M perforated, to manufacturer's recommendation
Drywall screws	ASTM C1002: Type S, self-tapping
Hangers	4 mm (8 gauges) galvanized wire
Tie Wire	1.5 mm (16 gauges) galvanized
Powder-actuated Anchors	Fed. Specifications A-A-1922A, A-A-1925A, A-A-55614
Masonry Nails	Hardened steel, screw type, 3.8 mm (9 gauge)

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Gypsum Wallboard General Purposes	ASTM C1396 / C1396M, Type X, one-hour fire-rated (Gypsum Board alone). Regular wallboard shall be 12.5mm or 16mm thick gypsum panels, as required by application, to COMPANY approval with tapered edges (minimum 1200 mm by 2400 mm). Specification to be agreed with COMPANY per application
Gypsum Wallboard Moisture Resistant	ASTM C1396 / C1396M, Type X, one-hour fire-rated (Gypsum Board alone). Moisture resistant wallboard shall be 12.5mm or 16 mm thick gypsum panels, as required by application, to COMPANY approval with tapered edges (minimum 1200 mm by 2400 mm). Specification to be agreed with COMPANY per application
Laminating Adhesive	As recommended by gypsum wallboard manufacturer
Fasteners	Type and size recommended by gypsum drywall manufacturer for the substrate and application indicated
Trim accessories	Provide corner beads, edge trim, and control joints, which shall comply with ASTM C1047
Acoustical Sealant	Water base, non-drying, non-bleeding, non-staining type and permanently elastic or as recommended by the gypsum board manufacturer for application
Insulation (Exterior Walls)	Insulation for dry-lining to external walls shall be to the requirements of AGES-SP-01-017 Masonry Wall and Thermal & Moisture Protection, as approved by COMPANY

6.3.3 Execution

6.3.3.1 Site Conditions

a) The temperature in areas to receive wallboard shall be maintained at a minimum of 13°C from 24 hours prior to start of installation until installation is complete and jointing materials are thoroughly dry.

b) Adequate ventilation shall be provided to eliminate excessive moisture during and following joint treatment applications.

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c) Environmental conditions (temperature, humidity and ventilation) shall be within limits recommended by manufacturer for optimum results.

6.3.3.2 Partition Framing

a) Unless otherwise specified, installation of framing members shall conform to the requirements of ASTM C754. Top and bottom tracks shall be accurately aligned in accordance with the partition layouts and secured to concrete with powder-actuated anchors or stub nails at 600 mm centres. Top tracks in contact with steel joists shall be wire tied or secured with clips. Studs shall be vertical, spaced at not more than 400 mm on centres. Sufficient clearance shall be maintained at the top of studs to provide for a possible 13 mm deflection. All studs adjacent to door frames, partition intersections and corners, and at dissimilar construction shall be secured to tracks with screws through each stud and track flange per MANUFACTURER'S instructions.

b) Stud splices, when necessary, shall be lapped 200 mm with two screws in each flange. Double studs shall be provided at all openings.

6.3.3.3 Furring

a) Furring channels shall be vertical, spaced not more than 600 mm on centres, and secured with powder-actuated anchors or masonry nails drive through alternate flanges into mortar joints or concrete walls at 400 mm centres. Horizontal furring channels shall be installed at the floor and ceiling line, and as required for proper support. Furring channels shall be plumbed and aligned by shimming.

b) Furring channels or screw studs shall be secured to steel columns as indicated or as required for UL fire rating.

6.3.3.4 Gypsum Board Installation

a) Unless otherwise specified, installation shall conform to the requirements of ASTM C840.

b) Wallboard shall be of maximum practical length to reduce the number of end joints. Ends and sides shall be in contact, but not forced into place. Wallboard shall be cut neatly to fit around all outlets, switch boxes, recessed light fixtures, diffusers, and similar items.

c) On walls and partitions, gypsum wallboard shall be installed with long dimension vertical. All side joints shall be made over framing members. Joints on opposite sides of partitions shall occur over different studs. End joints, if required, shall be neatly fitted and staggered. Boards shall be fastened to each framing member with screws spaced not more than 300 mm apart in the field and edges of boards. Space screws 300 mm on center in field of panels and 200 mm on center staggered along vertical abutting edges.

d) Use maximum lengths to eliminate end joints. Fit edges closely, but do not force together.

e) On walls and partitions, moisture resistant wallboard where specified on the drawings, shall be installed with long dimension horizontal. All end joints shall be neatly fitted and staggered and shall be made over framing members. Joints on opposite sides of partitions shall occur over different studs. Cut edges and openings shall be sealed as recommended by the wallboard manufacturer. Boards shall be fastened to each framing member with screws spaced not more than 200 mm apart in the field and ends of boards.

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f) Walls or partitions to be double course shall have second layer of gypsum board laminated with adhesive or screwed over base layer. Installation shall be in strict conformance with specified handbook procedures or gypsum board manufacturer's recommendations.

g) Screws shall be driven not less than 10 mm, or more than 15 mm, from ends and edges of boards. All screw heads shall be dimpled slightly below the surface of the board, with particular care taken not to break the face paper or otherwise damage the surface or core of the board.

6.3.3.5 Insulation

Where specified, insulation shall be installed tightly abutting furring channels around the entire perimeter. All joints shall be sealed with an approved gun-grade mastic sealant compatible with the insulation. Acoustical sealant shall be applied to the perimeter of the framing prior to the installation of the gypsum board.

6.3.3.6 Accessories

a) Corner beads shall be installed at all external corners, horizontal and vertical.

b) Metal trim members shall be installed wherever gypsum board abuts other materials, and on exposed edges of gypsum board.

c) Expansion joints shall be installed at a maximum of 10 meters unbroken run.

d) Control joints shall be provided in walls at any break in a continuous run for doors or windows. Control joints shall be installed at jambs, from the opening header to the ceiling or structure above.

e) Install and fasten corner beads with staples 230 mm O.C. on both flanges along entire length of bead. Install metal trim around openings, where wallboard drywall terminates, and at other dissimilar material. Fasten with screws or staples spaced 300 mm O.C.

f) Power drive screws with an electric screwdriver so screw heads make a slight depression below surface of interior wallboard without breaking face paper. Drive screws no closer than 10 mm from edges and ends of board.

6.3.3.7 Finishing

a) Where gypsum wallboard is to be painted or is to receive vinyl fabric wall covering, joints, fastener dimples, and other depressions shall be finished with tape and joint compound, with joint compound feathered and sanded smooth. Wallboard surfaces shall be left smooth, clean, and ready for painting or the application of the wall covering.

b) Where gypsum wallboard is to be covered with ceramic tile, joints shall be finished with tape and joint compound, with joint compound feathered and trowelled smooth. No sanding will be required. Dimples and depressions need not be filled. Special waterproofing joint compound as recommended by the manufacturer shall be used in shower and drying areas. Wallboard surfaces and edges of cut-outs shall be sealed, and openings around pipes and fixtures shall be caulked flush. Wallboard surfaces shall be left smooth, clean, and ready for the application of ceramic tile.

c) Apply treatment at gypsum board joints (both directions), flanges of trim accessories, penetrations, fastener heads, surface defects and elsewhere as required to prepare work for decoration. Prefill open joints and rounded or bevelled edges, if any, using type of compound recommended by manufacturer.

d) Apply joint tape at joints between gypsum boards, except where trim accessories are indicated.

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e) Apply joint compound in 3 coats (not including prefill of openings in base), and sand between last 2 coats and after last coat.

f) Apply skim coat plaster to metal stud partition system, in office and personnel rooms, and to other rooms as indicated on the drawings.

6.3.3.8 Clean-up

a) Floors and other surfaces shall be cleaned of all jointing compound droppings and all rubbish and debris resulting from the work shall be removed.

b) Suitable protection measures shall be applied / considered to ensure that there is no damage or deterioration to gypsum board assemblies until the construction is complete and the whole system / building is handed over to COMPANY.

6.4 Tiles

6.4.1 General

6.4.1.1 Scope of Work

a) This section covers the supply of materials and installation for ceramic mosaic floor tiles, glazed ceramic floor & wall tiles, quarry floor tiles, acid resistant ceramic floor & wall tiles, porcelain stoneware tiles, cast marble tiles, Terrazzo and precast roof tiles.

b) Tiling shall be installed where indicated as per finishes schedule.

6.4.1.2 Related Work Specified Elsewhere:

a) See AGES-SP-01-017 Masonry Wall and Thermal & Moisture Protection.

b) See Gypsum Drywall, Section 6.3 herein.

c) AGES-SP-01-013 Architectural Design Basis, see 'APPENDIX A - Typical Finishes Schedule'.

6.4.1.3 Submittals

a) Manufacturer's product information including adhesive, grout, sealant, etc. installation requirements and other data as may be required to certify compliance with performance requirements. System and installation shall be in accordance with the requirements of Architectural Design Basis AGES-SP-01-013.

b) The following shall be submitted:

i. Shop drawings indicating tile patterns if applicable and locations and widths of control and expansion joints in tile surfaces

ii. Source of Materials: Provide materials obtained from one source for each type and color of tile, grout and setting materials

iii. Test Report: Sample Test Report that indicates the compliance of the product with the requirements of specified codes and standards

iv. Minimum 2 nos. of each selected samples shall be submitted to the COMPANY, after the colors have been selected. Color shall be selected by the COMPANY from the MANUFACTURER's

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standard color range. Submit 300 mm (12-inch) by 300 mm (12-inch) sample panel showing each type and size ceramic tile with grout in color as selected

v. VOC Classification: CONTRACTOR shall submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013

c) Ceramic mosaic tile samples shall be suitably mounted and identified. At least two samples representing each pattern selected shall be submitted.

d) Glazed tile and quarry tile samples shall be full size and shall be suitably mounted, shall be marked with an identifying color name and the manufacturer's designation, and shall represent the full color range of the material. At least two samples shall be submitted in each color selected. Each sample board shall consist of no fewer than four individual tiles and shall show grouted joints and selected pattern.

e) Marble, porcelain stoneware and cast marble tile samples shall be a full size single tile, marked with an identifying color name and the manufacturer's designation. At least two samples shall be submitted in each color and finish selected.

6.4.1.4 Handling and Storage

All materials shall be packaged at the factory and shipped to the project site in unbroken grade-sealed containers. Material containers shall be unopened and shall be clearly marked with the manufacturer's name, brand and color coded grade seal. Materials shall be carefully handled and stored in weatherproof enclosures.

6.4.1.5 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).
b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

6.4.2 Materials

6.4.2.1 Materials shall be as follows:

a) Materials shall conform to the following standards and quality levels. Products will be considered upon submittal reviews and approval by the COMPANY.

Ceramic Tile	Shall conform to ANSI A137.1, porcelain, standard grade, cushion edge, back mounted, with trim shapes unglazed. Minimum thickness shall be 10 mm
Glazed Wall Tile	Shall conform to ANSI A137.1, glazed tile with cushioned edges and trim edged with lead free finish. Minimum thickness shall be 6 mm

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Acid Resistant Tiles	Shall be heavy duty wear and chemical resistant tiles and shall be tested to BS EN ISO 10545 or ASTM C650. Minimum thickness shall be 10 mm
Quarry Tile	Shall conform to ANSI A137.1, unglazed quarry tile and trim with smooth surface, factory polished quarry tile shall contain a minimum of 17% post-industrial recycled content. Minimum thickness shall be 12 mm
Marble	Shall conform to ASTM C503 / C503M. Marble shall include stone that is sawed, cut, split, or otherwise finished or shaped into blocks, slabs or tiles, and shall specifically exclude moulded, cast and artificially aggregated units composed of fragments, and also crushed and broken stone. Minimum thickness of marble tile shall be 20 mm
Porcelain Stoneware Tile	Unglazed / glazed porcelain tile and trim with the color extending uniformly through the body of the tile. Minimum thickness shall be 10 mm
Cast Marble Tile	Cast Marble or cultured Marble tile material usually composed of marble aggregates in a resin binder and pigments. Minimum thickness shall be 12 mm
Terrazzo Tiles	Factory ground and polished, square / chamfered edges. Minimum thickness shall be 20 mm
Granite Tiles	Factory ground and polished, square / chamfered edges. Minimum thickness shall be 20-30 mm
Concrete roofing tiles	Concrete roofing tiles shall be flat tiles 50 mm thick and 500 mm square
Portland Cement	ASTM C150, Type I
Sand	ASTM C144 or BS EN 13139, clean natural sand
Hydrated Lime	ASTM C207, Type S
Dry-set Mortar	ANSI A118.1, bearing TCNA seal. As recommended by tile manufacturer to COMPANY approval
Organic Adhesive	ANSI A136.1, Type I as recommended by tile manufacturer to COMPANY approval
Waterproofing Membrane	As recommended by tile manufacturer to COMPANY approval, 750 µm thick

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Waterproofing Membrane and Adhesive	One component polyurethane as recommended by tile manufacturer to COMPANY approval
Water	Clean and free from deleterious substances.
Ceramic and Quarry Tile Joint Filler	As recommended by tile manufacturer to COMPANY approval, colored where required by COMPANY
Cellular Backup	Polyethylene foam or closed cell butyl or neoprene rod, diameter 6 mm greater than nominal joint width
Expansion Joint Primer	As recommended by sealant manufacturer
Sealant	Multi component, gun and pouring grade, polysulphide sealant, conforming to ASTM C920.
Quarry Tile Edge Angles	Aluminum
Floor Sealer (for unglazed tile)	As recommended by tile manufacturer to COMPANY approval
Floor Tile Cleaner	As recommended by tile manufacturer to COMPANY approval
Waterproof Paper	Fed. Spec. UU-B-790A, Type I Grade C, Style 4. As recommended by tile manufacturer to COMPANY approval
Grout color	If advised by the COMPANY, approved proprietary brands of grout may be used in strict accordance with the manufacturer's technical specifications and recommendations

b) Supply an amount equal to 1.5 percent of each type, size, color, and surface finish of tile installed as spare.

6.4.2.2 Fabrication

a) Provide materials obtained from only one source for each type of tile and color to minimize variations in appearance and quality.

b) Ceramic tiles shall be provided in 200 mm x 200 mm / 300 mm x 300 mm size or other available size as required or as approved by the COMPANY and shall be laid in a square pattern. Mosaic tile shall be unglazed. Special trim units shall be provided at discontinuous edges, external and internal corners and bases.

c) Glazed wall tile shall be provided in nominal 200 mm x 200 / 300 mm or other available size as required or as approved by the COMPANY and shall be laid in a square or rectangular pattern. Special trim units shall be provided at discontinuous edges, external and internal corners, and bases. Wall tiles to be applied over full height of wall where indicated on drawings.

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d) Quarry tile shall be provided in 200 x 200 or 300 x 300 mm size and shall be laid in a square pattern. Special trim units shall be provided at external and internal corners, bases, tops of stairs, and elsewhere as required.

e) Marble stone tile shall be provided in 600 x 600 mm, 300 x 300 mm and other sizes as indicated on the drawings. Pattern shall be laid as shown on drawings. Finishes shall be as indicated in the finish schedule. Edge treatments shall be as detailed in the drawings. Granite for staircase or steps at entrance shall be used subject to COMPANY approval.

f) Cast marble tile shall be provided in 600 x 600 mm, 300 x 300 mm and other sizes as indicated on the drawings. Pattern shall be laid as shown on drawings. Finishes shall be as indicated in the finish schedule. Edge treatments shall be as detailed in the drawings.

g) Marble thresholds shall be double bevel, Grade A Italian marble or an approved substitute.

h) Setting materials to marble thresholds shall be material as recommended by the tile manufacturer with waterproof epoxy tiling grout and incorporating an anti-fungicide additive.

i) Porcelain stoneware tile shall be provided in 400 x 400 mm, 300 x 300 mm and other available sizes as required. Pattern shall be laid as shown on drawings. Finishes shall be as indicated in the finish schedule. Edge treatments shall be as detailed on the drawings.

j) Floor tile shall be 600 mm by 600 mm maximum size, impervious unglazed, non-slip ceramic tile with all-purpose edges. Colors and patterns shall be as selected. Base trim pieces shall match floor tile in size and pattern. Provide tile on factory-mounted sheets.

k) Skirting tile shall be 600 mm by 150 mm maximum size, impervious unglazed, ceramic tile with all-purpose edges. Colors and patterns shall be as selected. Base trim pieces shall match floor tile in size and pattern. Colors and patterns shall be as selected.

l) Trim and Special Shapes shall be provided as required:

   i. Rounded external corners, and trim shapes at head, jamb and sills of opening, of same material and finish as field tile, and as follows:

   ii. Wainscot Cap: Surface-bullnose or bullnose cap

   iii. External Corners: Bullnose shapes, unless otherwise shown

   iv. Internal Corners: Field-butted square

m) Tile setting adhesive/waterproofing shall be Ultra-Set polyurethane / tile setting adhesive as recommended by the tile manufacturer.

n) Grouting materials shall be commercial white sanded latex Portland Cement type for walls and epoxy grout for floors. Premixed grout may be used.

o) Terrazzo tiles consist of marble, granite, onyx or glass chips in Portland Cement, polyacrylate modified Portland Cement, or resinous matrix binder. The Terrazzo is poured, cured, ground, and polished. Terrazzo is typically used as a finish for floors, stairs or walls. Tile shall be provided in 300 x 300 mm, or size & pattern, etc. as indicated on the drawings. Tiles shall be rectified on all four sides and edges to be chamfered. Edge treatments shall be as detailed in the drawings.

p) Chemical / Acid resistant tiles shall be used for rooms where high chemical attack can be expected, e.g. battery rooms, laboratories, etc., Tile shall be provided in 200 x 200 mm / 300 mm by 300 mm maximum size, minimum 10 mm thick, non-slip ceramic tile with resin based mortar as recommended by selected manufacturer. Colors and patterns shall be as selected, or as indicated

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on the drawings. The tiles as specified above and with high resistance to chemicals shall be installed. Testing shall be done in accordance with BS EN ISO 10545. Size and type shall be agreed with COMPANY.

q) Colors and finishes will be selected by the COMPANY from the manufacturer's complete line of colors after the award of contract.

6.4.3 Execution

6.4.3.1 Surface Preparation

a) Surfaces to be covered with tile shall be cleaned of all dirt, dust, plaster, oil, grease, wax, paint, or other foreign substances. Surfaces to receive a mortar bed shall be kept saturated for 24 hours before mortar is placed. If the concrete slab does not absorb water, or if a chemical curing compound has been used on the slab, the entire concrete surface shall be scarified with a dust free cleaning system which requires no liquids or chemicals. Surfaces to receive dry-set tile shall be dry and shall not vary from a true plane more than 3 mm in 2,400 mm for walls and 3 mm in 3,000 mm for floors. Surfaces to receive tile set in organic adhesive shall be dry and shall not vary from a true plane more than 3 mm in 2,400 mm for walls and 2 mm in 900 mm for floors.

b) Installer must examine the substrate and the conditions under which ceramic tile is to be installed and notify CONTRACTOR in writing of conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions have been corrected in a manner acceptable to Installer.

c) If concrete block wall surface variations exceed 3 mm in 2,400 mm, the walls shall receive a levelling coat at least 13 mm thick. The levelling coat shall be mixed in the proportions of 1 part Portland cement, 1/2 part lime, and 5 parts damp sand.

6.4.3.2 Wall Tile Installation

a) This section specifies to comply with applicable parts of ANSI A108 series of tile installation standards and follow TCNA 'Handbook for Ceramic Tile Installation'.

b) Installation method related to subsurface wall conditions, grout types, etc. and to be selected as follows.

c) Masonry Exterior: TCNA W202 -7.

d) Masonry Interior: TCNA W212 (with levelling coat).

e) Masonry, Interior: TCNA W213.

f) Metal Studs, dry walling, interior: TCNA W244.

g) Tile shall be carefully laid out and centred on each wall or section of wall to avoid small cuts. Each cut tile shall be no less than one half its original dimension. Cut tile edges shall be rubbed smooth. Joints shall be approximately 2 mm wide and accurately aligned. All exposed and discontinuous edges shall be laid with bullnose cap or edge units. All external corners shall be laid with bullnose corner units. All vertical internal corners shall be laid square. All horizontal internal corners shall be laid with cove units. Tile shall be installed over the flashing flanges of service sinks and tight to the surfaces of the sink curbs. Finished tile surfaces shall be smooth, even, and free from offsets or irregularities, with straight, uniform joints.

h) Install ceramic tile work in accordance with the recommended systems by the tile manufacturer.

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i) Extend tile work into recesses and under equipment and fixtures, to form a complete covering without interruptions, except as otherwise shown. Terminate work neatly at obstructions, edges, and corners without disruption of pattern or joint alignment.

j) Fit tile closely to electrical outlets, piping, and fixtures so that plates, collars, or covers overlap tile.

k) Set marble threshold in same type setting bed as field tile.

6.4.3.3 Dry-Set Method for Wall Tile

a) Materials and workmanship shall be in accordance with ANSI A108.5 and TCNA Method W202.

b) Dry-set mortar shall be mixed in accordance with the manufacturer's instructions. Additional mortar, water, or other ingredients shall not be added after the initial mixing period. The mortar shall not be used after it has begun to set.

c) The surfaces to be covered may be dampened but shall not be saturated. The dry-set mortar bed shall have a thickness of at least 2.5 mm. The mortar shall be floated over an area no larger than can be covered with tile while the mortar remains plastic. The mortar shall be evenly spread and combed with a notched trowel not more than 10 minutes before placing tile.

6.4.3.4 Dry-Set on Levelling Coat

a) Materials and workmanship shall be in accordance with ANSI A108.1 and TCNA Method W211.

b) A levelling coat of mortar, 13 mm thick minimum, shall be applied where necessary, to the masonry wall surfaces to be covered with tile. The levelling coat shall be allowed to cure for a minimum of 20 hours at 20°C or above prior to tile placement. Dry-set mortar shall be mixed in accordance with the manufacturer's instructions. The prepared dry-set mortar bond coat, 2.5 mm thick minimum, shall be applied over the cured levelling coat. The surface to be covered may be dampened but shall not be saturated. The bond coat shall be floated over an area no larger than can be covered with tile while the bond coat remains plastic. The bond coat shall be evenly spread and combed with a notched trowel not more than 10 minutes before placing the tile.

c) The tile shall be thoroughly embedded in the bond coat while the bond coat is still plastic. The surfaces of the tile shall be uniform with edges of adjacent tile at the same plane.

d) While the bond coat is still plastic, if the COMPANY so requests, the CONTRACTOR shall remove one or more tile units to verify embedment. If any tile is found to be improperly bedded, the COMPANY may require a part or all of the tile to be removed and re-laid.

6.4.3.5 Organic Adhesive

a) Materials and workmanship shall be in accordance with ANSI A108.4 and TCNA-212-07 Method W223.

b) Organic adhesive is a prepared organic material for interior use only, ready to use with no further addition of liquid or powder, which cures or sets by evaporation.

c) Organic adhesives are suitable for thin-setting tile on floors, walls, and countertops, where surfaces are appropriate and properly prepared in accordance with adhesive manufacturers' directions.

d) Surfaces to be covered shall be clean and dry. Adhesive shall be spread with a notched trowel so that the entire surface is covered with no bare spots. Adhesive that films over before tile is applied shall be removed and replaced with fresh adhesive.

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e) Adhesives shall be applied under conditions and temperatures recommended by the MANUFACTURER and shall be verified for the possibility of application in hot weather.

6.4.3.6 Floor Tile

a) All floor tiles shall be laid by the dry-set method with the exception that quarry tile shall be laid by the dry-set method on a cured mortar bed.

b) Ceramic tile joints shall be approximately 1-2 mm wide, to COMPANY approval. Quarry tile joints shall be uniform width, 10 mm wide. All joints shall be maintained in accurate alignment. The tile work shall be carefully laid out and centred to avoid small cuts. Each cut tile shall be no less than one half its original dimensions. All cuts shall be rubbed smooth and even.

6.4.3.7 Dry-Set Methods for Floor Tile

a) Materials and workmanship shall be in accordance with ANSI A108.5 and TCNA Method F113.

b) Dry-set mortar shall be mixed in accordance with the manufacturer's instructions. Additional ingredients shall not be added after the initial mixing period. The mortar shall not be used after it has begun to set.

c) The surfaces to be covered may be dampened, but not saturated. The dry-set mortar bond coat shall have a thickness of at least 2.4 mm. The mortar shall be floated over an area no larger than can be covered with tile while the mortar remains plastic. The mortar shall be evenly spread and combed with a notched trowel not more than 10 minutes before placing tile.

6.4.3.8 Dry-Set on Mortar Bed

a) Materials and workmanship shall be in accordance with ANSI A108.6 and TCNA Method F132.

b) Quarry tile shall be set in acid resistant mortar.

c) Mortar for the mortar bed shall be mixed in the proportions of one part Portland cement and six parts damp sand, by volume. Mortar shall be machine mixed and shall be low slump. Mortar shall be mixed and placed only as required by the tile setter. Mortar which has reached its initial set shall be discarded. No remixing with additional water, cement, or other materials will be permitted.

d) The mortar bed shall be placed over saturated concrete, tamped, and screed to a thickness of 20 mm minimum. High areas of the concrete slab shall be ground to allow for proper mortar thickness. Before setting tile, the mortar bed shall be allowed to cure for a minimum of 20 hours at 20°C or above.

e) Tile shall be set in a bond coat of acid resistant mortar. The bond coat shall be mixed in accordance with the manufacturer's instructions and placed only as required by the tile setter. Bond coat which has reached its initial set shall be discarded. No remixing with additional water will be permitted.

f) Total thickness of the bond coat between tile and the cured bed shall be approximately 2.5 mm.

6.4.3.9 Epoxy Mortar

a) Epoxy mortar is suitable for thin-set installations of ceramic and stone tiles where chemical resistance of floors, high bond strength, and high impact resistance are important considerations. When specified surface levelness or uniformity of slope is obtained, prepare floors for setting bed in accordance with TCNA Handbook. Floors shall be free from sealers, coatings, oil, dirt, and dust. Prepare floors before application of resin mortar in accordance with printed instructions and

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recommendations of the mortar manufacturer. Using a plain (not serrated) trowel, apply a continuous setting bed of chemical-resistant mortar. Place tile into setting bed and tap lightly to a true plane. Install grout and tile with water cleanable tile setting and epoxy grouting.

b) Complete installation and material specifications are as contained in ANSI A108.6 and ANSI A118.3.

6.4.3.10 Liquid Waterproofing

Tile installed in wet areas shall be set in liquid applied waterproofing. Liquid applied waterproofing bond coat shall be evenly spread on the floor surface using a notched trowel. Application procedures shall be in accordance with the manufacturer's instructions. Any material which begins to set before tile placement shall be removed and replaced.

6.4.3.11 Tile Setting

a) Ceramic tile and quarry tile shall be set while the bond coat is still plastic. Tile shall be pressed or beaten into the bond coat until there are no voids beneath the tile. Quarry tile shall be placed against a metal straightedge with all grooves in the same direction.

b) At discontinuous exposed edges, tile shall be set to match the finish surface elevation of adjacent floors.

c) After the tile is laid, the entire floor shall present a true and even surface varying not more than plus or minus 3 mm from a 3000 mm straightedge.

d) While the mortar is still plastic, if the COMPANY so requests, the CONTRACTOR shall take up one or more tile units to verify embedment. If any tile is found to be improperly bedded, the COMPANY may require a part or all of the tile to be removed and re-laid.

6.4.3.12 Joint Filling

a) When the bond coat or adhesive has hardened sufficiently to lock the tile in place, joints shall be saturated with water, then filled with joint filler, and finished smooth and flush with the surface of the floor.

b) Color of joint filler will be selected by the COMPANY from the manufacturer's standard colors.

c) Joint filler shall be spread over the tile surface until joints are filled.

d) Placing of joint filler in the joints shall be carefully done to ensure that each joint space is completely filled to the bottom and that the filler is pressed into the joint to force it into firm contact with bedding mortar.

e) All surplus material shall be removed before it has hardened.

6.4.3.13 Expansion Joints

a) Expansion joints shall be provided at a maximum distance of 5.0 m in each direction and installed in accordance with TCNA Method EJ171 and as specified herein.

b) Locate expansion joints and other sealant filled joints, including control, contraction isolation joints, prepare joints and apply sealants to comply with requirements of referenced standards and sealant manufacturer or in the following manner.

   i. Edges of expansion joint grooves shall be cleaned by surfacing them with an abrasive disk, then blowing out the residue with a jet of air

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ii. The bottom part of each groove more than 15 mm deep shall then be filled with cellular backup material to within 13 mm of the top surface of the tile

iii. Edges of the grooves (except metal surfaces which shall not be primed) shall then be primed with expansion joint primer

iv. The remaining 13 mm of the joint, or all of joints not more than 15 mm deep, shall then be filled to within 1.5 mm of the top with Thiokol sealant

c) Sealant shall be mixed and applied in accordance with the manufacturer's recommendations. All sealants shall be approved by COMPANY prior to installation. Products by other manufacturers which are of equal quality and performance shall be submitted for COMPANY review and subject to approval.

6.4.3.14 Parting Strips or Transition Strip

Where the same floor covering does not extend through a doorway, a parting strip or transition strip shall be set to cover the edge of the floor covering or parting line between floor coverings. Except as otherwise indicated on the drawings, such parting strips or Transition Strip shall lie below and be centred on the door when it is closed. Parting strip type shall be approved by COMPANY.

6.4.3.15 Joints

Jointing Patterns: Unless otherwise shown, lay tile in grid pattern. Align joints when adjoining tiles on floor, base, walls and trim. Lay out tile work and centre tile fields in both directions in each space or on each wall area. Adjust to minimize tile cutting. Provide uniform joint widths, unless otherwise shown.

6.4.3.16 Cleaning and Protection

a) Upon completion of the various parts of this work, the CONTRACTOR shall remove all surplus material and rubbish.

b) As soon as the joint filler has set sufficiently, but while the surface is still damp, floors shall be covered with a continuous membrane of waterproof paper to prevent rapid drying and to protect the surface. All seams in the membrane shall be lapped at least 100 mm and tightly sealed. The membrane shall be left in place until all danger of marring has passed and the floors are ready for final cleaning, but in no case less than 5 days.

c) Clean tile surfaces as thoroughly as possible on completion of grouting.

d) Remove grout haze, observing tile MANUFACTURER'S recommendations about the use of acid and chemical cleaners.

e) Rinse tile work thoroughly with clean water before and after using chemical cleaners.

f) Polish surface of tilework with soft cloth.

g) Prohibit foot and wheel traffic on newly tiled floors for at least 3 days, preferably 7 days. Place large, flat boards in walkways and wheelways for 7 days where use of new floor is unavoidable.

h) Wall surfaces shall be protected from rapid drying and shall be kept moist for at least 72 hours after the tile is applied.

i) After all work has been completed and just before final inspection, protective coverings shall be removed and each floor surface shall be inspected and any defects repaired.

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j) Tile surfaces shall be thoroughly cleaned with the specified tile cleaner and rinsed completely with clear water. Factory applied wax shall be removed from quarry tile surface.

k) Unglazed tile shall be sealed with the specified floor sealer in accordance with the procedures recommended by the manufacturer.

6.5 Acoustical Tile Ceiling

6.5.1 General

6.5.1.1 Scope of Work

a) This section covers lay-in acoustical panel ceilings and decorative metal ceiling systems with concealed grid suspension systems. Ceiling works shall be state-of-the-art, functional and complete in all parts in accordance with this Specification.

b) The CONTRACTOR shall be responsible for the integration of the various discipline requirements i.e. (Mechanical systems, Electrical systems, HVAC, lighting, fire detection etc.) and shall take into account the location of diffusers, lighting fixtures, plumbing works, fire and gas systems and sprinklers etc.

c) In principle, ceiling panels shall be removable offering to access to the ceiling void for routine inspection and maintenance of services.

d) For each type of ceiling, representative prototypes shall be submitted to the COMPANY for approval.

e) Suspended ceiling shall be installed where indicated as per the finishes schedule.

f) The proprietary suspended acoustic ceilings shall be fitted flush around light fittings, ventilation grills, etc. and shall conform to the following requirements:

   i. Shall be non-combustible

   ii. Shall not allow the spread of flames

   iii. Shall be vermin proof

   iv. Shall easily be cleaned or redecorated

   v. Shall be so designed as to allow the easy replacement of panels

   vi. Shall provide the sound absorption coefficients to maintain the acceptable noise / sound level of the room

6.5.1.2 Related Work Specified Elsewhere:

a)	Electrical Engineering Design Guide	AGES-GL-02-001
b)	HVAC Equipment	AGES-SP-14-002
c)	HVAC Ductwork, Pipework and Accessories	AGES-SP-14-003
d)	Building Safety Specification	AGES-SP-03-003

e) AGES-SP-01-013 Architectural Design Basis APPENDIX A – 'Typical Finishing Schedule'

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6.5.1.3 Submittals

a) Manufacturer's specifications, installation instructions and product data on metal grid system components, acoustical units, and all other products to be used. System and installation shall be in accordance with the requirements of Architectural Design Basis AGES-SP-01-013.

b) Samples:

i. Submit 600 mm (24-inch) by 600 mm (24-inch) sample panel and showing each piece of suspension system members, moulding and sound insulation. Color shall be selected by the COMPANY from the Manufacturer's standard colors.

ii. Submit three samples of acoustic materials to illustrate color and range of appearance exposed materials including the following.

iii. Grid system components, including suspension system main runner, cross runner, edge trim, and all special shapes, in 12 inches (300 mm) lengths.

c) Shop Drawings: Show grid layout and dimensioning, panel layouts, lighting fixtures, air diffusers, grilles, and all other items exposed in acoustical ceilings, Show details of interfaces with other work or ceiling finishes, and special conditions if any.

d) Warranty: Manufacturer's standard warranty for extent of conditions allowable and duration of warranty.

e) VOC (Volatile Organic Compound) Classification: CONTRACTOR to submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013.

6.5.1.4 Handling and Storage

a) Deliver materials shall be original, unopened protective packages, with manufacturer's labels, legible and intact, indicating brand name, pattern, size, thickness, fire rating, etc.

b) The materials shall be stored in a dry and clean area enclosed and protected from the elements.

c) Storage time of materials at the job site should be as short as possible and environmental conditions should be as near as possible to those specified for intended use / functions.

d) Excess humidity during storage can cause expansion of material and possible warp, sag, or poor fit after installation. Chemical changes in the mat and / or coatings can be aggravated by excess humidity and cause discoloration during storage, even in unopened cartons. Cartons should be removed from pallets and stringers to prevent distortion of material. Long-term (6-12 months) storage under uncontrolled environmental conditions should be avoided.

e) Damaged or deteriorated materials should be removed from the premises. Immediately before installation, to stabilize tile and panels, store them at a location where temperature and humidity conditions are similar to those ambient during installation and anticipated for occupancy.

6.5.2 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

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6.5.3 Materials

a) Suspended ceiling acoustical panels shall be in accordance with ASTM E 1264 classifications, or equivalent standards, as designated by types, patterns, acoustical ratings, and light reflectance.

b) Ceiling Assemblies shall be fully accessible suspended modular grid type ceiling panels with concealed insulation pads where required and concealed threaded rod suspension system.

c) Materials shall conform to the following standards and quality level. Products will be considered upon submittal review and approval by the COMPANY.

Acoustical Panels	Conforming to ASTM E1264, Class A rated, Flame spread 25 or less; and classified with Underwriter's Laboratories label on product carton, all acoustical panels shall contain an antimicrobial treatment. Type to COMPANY approval
Size, Nominal	600 mm x 600 mm or as specified in the project drawings
Finish	Mineral fibre, textured and with factory applied washable vinyl paint
Metal Panels	Fire rated, micro perforated, acoustically insulated, shall be made up of pre-finished powder coated aluminum, with acoustic pad and minimum 50 mm thick glass wool type insulation. Type to COMPANY approval
Size, Nominal	600 mm x 600 mm x 0.7 mm (minimum) or as specified in the project drawing
Finish	Galvanized and factory coated or as specified in the finish schedule
Concealed grid suspension system (main runner and cross tees, and edge mouldings)	Conforming to ASTM C635 / C635M, intermediate duty system; components fabricated from commercial quality hot dipped galvanized steel complying with ASTM A653 / A653M. Beams and tees formed from commercial quality cold-rolled steel, hot dipped galvanized coated and pre-painted. Type to COMPANY approval
Hangers	2.6 mm (12 gauges) galvanized steel wire
Powder-actuated anchors	To COMPANY approval

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Masonry Nails	Hardened steel, 3.9 mm (9 gauge)
Connectors and Clips	As per Manufacturer's standard
Hanger and Bracing Wires	Conforming to ASTM A641 / A641M
Fasteners	Fasteners used for attachment of acoustical ceilings to building superstructure
Edge Sealer	Latex adhesive designed for the purpose of sealing field-cut edges of acoustic panels
Insulation Pads	Concealed by panels. Fill Material: Glass wool type Density: 30 kg/cubic meter for glass wool Covering: Fire-retardant black fabric (80 grams per square meter) Thickness: 30 mm, minimum

d) Panels shall be completely sealed with reinforced aluminum top sheet.

e) Spare materials equal to 1.5 percent of each type of acoustical tile and suspension system components shall be furnished for use in future maintenance work.

6.5.4 Execution

6.5.4.1 Installation

a) Suspended acoustical ceilings shall be installed in accordance with the best industry practice and the recommendations of the ceiling system manufacturer. Each area shall be permanently enclosed before ceiling work is started.

b) Suspension system shall consist of heavy duty extruded aluminum or hot-dipped galvanized steel track to the structural support by bolts or threaded rods.

c) Before proceeding with acoustical work installation, ensure that all the wet works are completed and surfaces are dry.

d) The installer shall examine substrates scheduled to receive suspended acoustical units for unevenness, irregularities, and dampness that would affect quality and execution of work under which this work is to be performed and notify COMPANY, in writing, of conditions detrimental to the proper completion of the work. Do not proceed with work until unsatisfactory conditions are corrected. Beginning work means installer accepts substrates and conditions.

e) Installation of Suspension System for panels shall strictly comply with MANUFACTURER'S instructions and recommendations, except where more restrictive requirements are specified in this section.

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f) Install ceiling panels in coordination with suspension system. Scribe and cut panels to fit accurately. Place materials to bear all around on suspension members.

   i. Non-Square Edge Panels: Rout and paint field cut edges as necessary to match factory edge

   ii. Orientation: Lay panels with grain (if any) running in one direction throughout the Project

   iii. Edging: Panels shall be installed with continuous powder coated Z-channel adjacent to all walls

g) Coordinate installation with other work to ensure proper location of related work such as light fixtures, mechanical fixtures, fire protection systems and the like.

h) The suspension systems shall be designed to support the ceiling, lighting fixtures, grilles, diffusers, and other required items without sagging. In each area, the suspension system shall be installed and levelled before acoustical panels or light fixtures are installed. Each grid system shall be levelled to within 3 mm in 3.5 m. The layout for each suspended acoustical ceiling shall be symmetrical about the centre lines of the space with equal borders on opposite sides and shall be square and parallel to the enclosing walls.

i) Hanger spacing shall not exceed 1200 mm, with additional hangers placed at each corner of each recessed light fixture and framed opening. Hangers shall be securely anchored to overhead construction. Expansion anchors or powder-actuated anchors in tension will not be permitted.

j) Main runner tees shall be spaced 1200 mm apart and supported with hanger wires not more than 1200 mm apart. Main runner tees shall be placed with lines parallel to the bordering walls. Locations and methods of splicing shall be as recommended by the manufacturer.

k) Main cross tees shall be spaced 1200 mm apart, fitted between main runner tees, and clipped thereto, as recommended by the manufacturer. Each line of main cross tees shall be straightened and aligned, and lines shall be perpendicular to main runner tees and parallel to each other.

l) Additional members shall be installed as required to frame and support all light fixtures, diffusers, grilles, and similar items.

m) Edge moulding shall be installed at all walls around the entire perimeter of each ceiling area, and around all columns. Prefabricated corner units shall be installed at all internal and external corners. Hemmed edges of mouldings shall be field-notched with a suitable tool at the exact point where tees fit in, for flush fit.

n) Acoustical panels shall be neatly cut to fit the spaces, with each panel supported on all sides by main runners, cross runners, or edge mouldings.

o) After erection, dirty or discolored exposed surfaces of the metal suspension system and ceiling panels shall be cleaned and left free of defects. Items which are damaged, or which cannot be properly cleaned shall be removed and replaced.

6.5.4.2 Protection

Protect the finished installation from damage during balance construction period. Remove any soiled or damaged items and replace with new before acceptance.

6.5.4.3 Adjusting, Cleaning

a) Adjust panels so that ceilings are in one plane and look uniform with no individual panels too high or too low.

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b) Clean exposed surfaces using materials and methods recommended by MANUFACTURER of material or product being cleaned. Remove and replace work that cannot be successfully cleaned.

6.6 Resilient flooring

6.6.1 General

6.6.1.1 Scope of Work

This Section covers vinyl-composition and vinyl tile floor coverings, rubber tiles, rubber stair treads, cove base and stair base. Provide resilient flooring as a complete, integrated system from one manufacturer consisting of flooring tile, adhesive, and ground devices. Flooring shall be applied where indicated as per the finishes schedule.

6.6.1.2 Related Work Specified Elsewhere:

a) See Company Specification for Architectural Specialties, Metal, Wood and Plastics

b) AGES-SP-01-013 Architectural Design Basis, see APPENDIX A - Typical Finishes Schedule

6.6.1.3 Submittals

a) After colors have been selected, samples of resilient floor covering materials shall be submitted to the COMPANY for examination.

b) Submit manufacturer's product specifications and installation instructions for each floor tile material required. Include manufacturer's recommendations for cleaning and maintaining tile, including precautions against material and methods that may be detrimental to the finishes of the units

c) For each finish product specified, two sets of each type, color, shade and finish of resilient flooring and accessory required, indicating color and pattern thickness, and manufacturer’s name. Base and reducing strip shall be submitted in 150 mm lengths

d) Flame Spread Certification: Submit manufacturer's certification that resilient flooring furnished for areas indicated to comply with required flame spread rating has been tested and meets or exceeds indicated or required standard

e) VOC (Volatile Organic Compound) Classification: CONTRACTOR to submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013.

6.6.1.4 Handling and Storage

Materials shall be delivered to the job in the original unopened containers clearly marked with manufacturer’s brand and name. Materials shall be handled carefully and stored in weatherproof enclosures. Materials shall be stored at not less than 20°C for 48 hours before installation.

6.6.2 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

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b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

6.6.3 Materials

a) The materials furnished shall conform to the following standards and quality level. Products will be considered upon submittal review and approval by the COMPANY.

Vinyl composition & Vinyl Tile	300 mm by 300 mm by 3 mm thick and shall conform to ASTM F1066 or BS EN ISO 10581, composition 1, Class 2 vinyl tile
Reducing Strip	Same material as floor covering formed into tapered strip
Edge Strip	Where resilient tile floor meets concrete floors, provide 3 mm thick vinyl or rubber tapered or bullnosed edge strips
Rubber Stair Treads	Type shall be to COMPANY approval - Square Nose, 6 mm (1/4 inch) thick at nose tapering to 3 mm (1/8 inch) at back, with two 25 mm (1 inch) abrasive strips recessed full tread width. Type per application, to be approved by COMPANY
Cove Base / Skirting	Fed. Spec. SS-W-40A, Style B; vinyl or PVC, minimum 2 mm thick. Same material as flooring may be considered, 100 mm high from top of flooring, laser cut from the flooring material with straight edge and mitred corners. Type per application, to be approved by COMPANY
Height	100 mm
Color	Selected by the COMPANY from the manufacturer's standard line of colors
Stair Base	Vinyl or rubber, 2 mm thick
Adhesives	As recommended by the floor covering manufacturer per BS EN 17668. Adhesive, primer, and other application materials shall be as recommended by the flooring manufacturer for specific installation conditions
Floor Levelling Material	Factory prepared, maximum thickness of 3 mm. All materials including underlayment / crack filler shall be as recommended by the flooring manufacturer

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Protective Paper	Fed. Spec. UU-B-790A, Type I, Grade C, Style 4; laminated paper with non-staining adhesive
Rubber tiles	Minimum 4 mm to 6 mm thickness, heavy duty type, style as indicated on the Finish Schedule. Material for substation switchgear floors shall be studded rubber flooring mat in accordance with COMPANY Specifications Provide approx. 100 mm high moulded coved rubber base where rubber flooring is specified. The base shall be complete with preformed external corners with returns of no less than 64 mm
Miscellaneous Materials	Furnish concrete slab primers, and other materials as recommended by the manufacturer of the resilient flooring
Cleaner and Polish	Floor cleaner and polish shall be as recommended by the flooring manufacturer
Built-in Special cleaning floor mat	Natural rubber / fibre, durable, washable, abrasion resistant and self-cleaning type, of elegant finish, color and pattern

b) Colors will be selected from the manufacturer's standard line of colors after the award of contract.

c) Spare material equal to 1.5 percent of tile used in each type / color shall be furnished for use in future maintenance work.

d) Where a resilient floor covering joints another type of floor finish at a doorway or other opening, or where a resilient floor covering joins another finish which has a different finish elevation, a parting strip shall be set to cover the parting line between the two floor finishes. Parting strips shall lie directly below and be centred on the door when it is closed. Parting strips shall be bronze and shall be equal, 6 mm by 28 mm where adjacent surfaces are the same elevation. Where adjacent surfaces are different elevations, suitable bronze strips 38 mm wide, tapered or otherwise formed to fit the elevation difference, shall be selected.

e) All floors indicated on the drawings as conductive areas shall be conductive and static dissipative vinyl tile. The tile shall be installed with conductive adhesive furnished by the tile manufacturer and installed in complete accordance with the manufacturer's installation instructions.

6.6.4 Execution

6.6.4.1 Surface Preparation

a) Inspect areas and conditions to receive flooring prior to installation. Do not start installation until unsatisfactory conditions are corrected. Installation of flooring shall be considered an acceptance of the substrate.

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b) The concrete subfloor scheduled to receive resilient flooring shall be broom cleaned, and then vacuum cleaned. The subfloor shall be free of hardening compounds, dust, scale, and other deleterious matter immediately prior to installation of flooring.

c) Installation of floor coverings shall not begin until the subfloor is completely dry and, in so far, as practicable, the work of all other trades has been completed. Before laying floor coverings, the floor surface shall be thoroughly cleaned of dirt, dust, plaster, oil, grease, paint, or other foreign substances. Holes, hollow spaces, and other imperfections in the surfaces to be covered with resilient floor covering shall be filled with floor levelling material. The levelling material shall cure for 24 hours before the floor covering is installed.

d) Use underlayment / crack filler to build up thickness as required to form a level and smooth base surface for resilient flooring. Underlayment shall be composition that can be smoothly applied and feathered out on edges. Material shall be latex cement and sand composition or other mastic composition as approved by the COMPANY. Materials shall be compatible with flooring adhesive. Apply latex type underlayment in single layers no greater than 3 mm thick and no more than 10 mm total thickness.

e) The floor shall be dry and shall be maintained at a minimum temperature of 20°C for 48 hours before floor covering is laid, during laying, and for 10 days after the work is complete.

6.6.4.2 Installation

a) Floor coverings shall be installed in compliance with BS 8203.

b) Do not lay the flooring until the concrete subfloor has thoroughly cured. Do not lay flooring sooner than 28 days after concrete has been cast. There shall be no exception to this rule. Perform required tests to determine that the concrete has cured.

c) Install resilient flooring materials and accessories after other Finishing operations, including painting, have been completed.

d) Install flooring in accordance with the MANUFACTURER'S written instructions.

e) Install tile and accessories in mastic cement conforming to the MANUFACTURER'S printed material. Room temperature shall be at least 21°C (70 degrees F) and shall be kept above that temperature for 3 days after installation.

f) Concrete primer shall be applied before adhesive, if recommended by the flooring manufacturer.

g) Install flooring in true, straight lines, closely fitted to adjacent flooring. Snap lines on both axes of the room, and space flooring from the center in both directions so that opposite margins will be uniform and no less than 1/2 tile in width. Where flooring abuts floors of greater thickness, ramp flooring with trowel type underlayment to meet adjacent floors in a flush joint.

h) Install resilient base in lengths as long as practical. Tightly adhere to backing throughout length.

i) Match tiles for color and pattern by using tile from cartons in same sequence as manufactured and packaged. Cut tile neatly around all fixtures. Broken, cracked, chipped, or deformed tiles are not acceptable.

j) Remove any excess adhesive or other surface blemish, using neutral tape cleaners as recommended by the flooring manufacturer.

k) Floor coverings shall be solidly cemented to the floor with adhesive applied in a continuous thin film and spread evenly with a notched steel trowel or other suitable application tool.

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l) Tile shall be laid with grain running in perpendicular direction between adjacent tiles.

m) Where floor coverings of different thicknesses abut, a tapered layer of latex levelling cement shall be installed under the thinner floor covering so the top surface of abutting pieces is at the same elevation.

n) Floor coverings shall be installed on stair treads and landings. Nose of rubber stair treads shall be trimmed to fit.

o) Except at doorways where metal thresholds are located, reducing strips shall be installed at exposed edges of floor covering where the elevation of the floor covering is higher than the adjacent floor surfaces.

p) Bases shall not be installed until the surfaces against which they are to be applied are dry and clean. Bases shall be set in accordance with the manufacturer's directions and continuously cemented in place. Joints and angles shall be neat and tight. Each base, throughout its entire length, shall have top and bottom edges in firm contact with walls and floors. Corners shall be prefabricated.

q) Stair base shall be scribed and cut to contour to fit tightly at each tread and each riser.

r) Rolls which have deformed because of improper storage shall be rejected.

6.6.4.3 Protection and Cleaning

a) Resilient floor coverings shall be protected from damage until final acceptance. Areas that are subject to traffic or over which materials or equipment are to be moved shall be temporarily covered with durable, non-staining, protective paper.

b) Just before final inspection, surfaces of resilient floor coverings shall be thoroughly cleaned, waxed, and polished with a heavy polishing machine. Chemicals and waxes used in cleaning and dressing floor surfaces shall be suited to the types of floor covering installed.

6.7 Carpeting

6.7.1 General

6.7.1.1 Scope of Work

a) This Section covers the installation of Modular Carpet tile or sheet carpeting. All carpet shall be of approved type, approved brand laid over Terrazzo / concrete flooring and treated with an approved sealant to reduce dust and pollution. Carpet tiles shall be of the same size so that they could be easily interchanged, relocated or replaced.

b) Carpet tiles or sheet carpeting shall be installed where indicated as per the finishes schedule.

6.7.1.2 Related Work Specified Elsewhere:

a) See Tile, Section 6.4 herein.

b) See Resilient Flooring, Section 6.6 herein.

c) Architectural Design Basis, AGES-SP-01-013 see 'APPENDIX A - Typical Finishing Schedule'.

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6.7.1.3 Submittals

a) Submit manufacturer's product literature, installation and maintenance instructions for carpeting material and installation accessory required. Products and installation shall be in accordance with the requirements of Architectural Design Basis AGES-SP-01-013.

b) Submit manufacturer's standard size samples showing full range of colors, textures and patterns available for each type of carpet tile required.

c) Submit Material Safety Data Sheet for adhesives, levelling compounds, and primers required.

d) VOC (Volatile Organic Compound) Classification: CONTRACTOR to submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013.

6.7.1.4 Handling and Storage

a) Carpet shall be delivered to site in original factory wrappings and containers, clearly labelled with identification of manufacturer, brand name, quality or grade, fire hazard classification and lot number or provide independent certificates.

b) Store materials in original undamaged packages and containers, inside well-ventilated area protected from weather, moisture, soiling, extreme temperatures, and humidity; laid flat, blocked off ground to prevent sagging and warping.

c) Maintain temperature in storage area as recommended by manufacturer.

6.7.2 Materials

a) Carpet tile shall be either all 'tops' or all 'bottoms' and shall be from the same dye lot. Arrows or other indicators shall be provided on the back side of the carpet tile to aid in installation. Tiles shall be installed with the grain running in the same direction for a monolithic appearance. Carpet tile shall be designed for free-lay installation.

b) Materials shall conform to the following standards and quality levels. Alternate products will be considered upon submittal review and approval by the COMPANY.

Carpet tiles / sheets	To COMPANY approval
Yarn Construction Surface	Level or textured loop, multi-level loop, or pattern
Yarn	Multi-ply
Finished pile thickness	Minimum 0.125 inch, maximum 0.5 inch (3.175 mm, maximum 12.7 mm)
Composition (Face yarn)	100 percent dyed Nylon, type to COMPANY approval
Effective Pile height	5.0 mm ± 0.5 mm
Pile weight	750 gm / square meter ± 10%
Total weight	4250 gm / square meter ± 10%

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Primary backing	100 percent synthetic materials, non-woven or as recommended by carpet manufacturer
Secondary Backing	Fiberglass reinforced secondary backing consisting of a polymeric cushion or hard back compound consistent with all performance testing requirements
Dye method	Yarn or solution dyed
Tile size	500 mm x 500 mm or as per carpet manufacturer standard
Tile total thickness	8.5 mm ±10%
Color / Pattern / Design	To be selected by the COMPANY
Flame spread	Pass critical radiant flux, Class I, a minimum critical radiant flux of greater than or equal to 0.45 watts/sq. cm when tested over fiber cement board according to ASTM E648
Smoke density	A maximum specific optical density of less than or equal to 450 based on when tested in the flaming mode according to ASTM E662
Static generation	3.5 kilovolts maximum at 20 percent relative humidity and 20oC in accordance with AATCC 134 for the useful life of the carpet tile
Wear warranty	10 year wear warranty (wear shall not exceed 10 percent of pile face fiber by weight for 10 years)
Anti-microbial treatment	Anti-microbial treatment Pass standards for inhibiting bacteria and fungus per AATCC 174, parts I, II and III. Lifetime warranted
Backing Integrity	No de-lamination for the useful life of the carpet tile
Soil / Stain Resistant	No apparent stain per AATCC 6, AATCC 122 and AATCC 175 for 10 years
Color Fast	No change in color due to exposure to light per AATCC 16E. No color change due to atmospheric contaminants, including ozone and / or oxides of nitrogen per AATCC 164 and AATCC 129. No stain by crocking per AATCC 165 (10 years)
Reducing Strip	As recommended by carpet manufacturer

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Cushion	Minimum 6 mm thick bonded polyurethane foam cushion, density 128 kg/m3 plus or minus 5 percent, anti-microbial, Compression Force Deflection (CFD) conforming to ASTM D-3574 at 65 percent equal 5625 kg/m2
Adhesive	Adhesives shall be free of synthetic latex, giving no volatile organic compound emission. Water-based adhesives or adhesive-free installation shall be used

c) Include other accessories as required by manufacturer for a complete installation.

d) Furnish an additional 1.5 percent of carpet to be stored on site for maintenance / replacement needs. Additional carpet shall be packaged for storage to prevent damage from sagging, warping, and other deleterious effects. Samples shall be marked with an identifying color name and the manufacturer's designation and shall represent the full color range of the material.

6.7.3 Execution

6.7.3.1 Installation

a) Installation of the carpet shall not begin until all other work in the areas to be carpeted is essentially complete. Floor surfaces over which the carpet is to be installed shall be cleaned of all dirt, dust, plaster, oil, grease, wax, or other foreign substances.

b) It shall also be level, smooth and free from scaling and irregularities.

c) Verify that surfaces are smooth and flat and are ready to receive work.

d) Verify concrete floors are dry to a maximum moisture content of 7 percent; and exhibit negative alkalinity, carbonization, or dusting.

e) Coordinate carpet installation with any under floor electrical / telephone / data systems such that necessary conductors and cables are pulled prior to carpet installation. Accessories (floor power / communications items, accessories) shall be installed on top of carpet after installation.

f) Remove subfloor ridges and bumps. Fill minor or local low spots, cracks, joints, holes, and other defects with subfloor filler.

g) Apply, trowel, and float filler to achieve smooth, flat, hard surface. Prohibit traffic until filler is cured.

h) Vacuum clean substrate.

i) Maintain room temperature prior to installation and relative humidity at approximately same as that at which the area is to be maintained or as per carpet manufacturer's instruction.

j) The carpet shall be accurately installed by skilled labor, as specified below, and in strict accordance with the manufacturer's instructions.

k) The centres of rooms or spaces shall be determined by establishing chalk lines that divide each room or space into quadrants. The resulting quadrants must meet at right angles. Centre chalk lines may be offset to ensure that perimeter carpet will be half size or larger.

l) To prevent movement of the carpet during initial stages of installation, releasable type adhesive shall be applied in strips alongside each centre chalk line. Carpet shall be laid accurately and firmly

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along the centre lines. Additional carpet tiles shall be free laid using a 'stair-step' procedure. To prevent carpet from shifting in larger spaces, additional adhesive shall be applied in strips on 10 meter centres maximum in both directions.

m) Carpet shall be tightly butted together with joints accurately aligned. Care shall be taken when butting carpet to avoid too much pressure, which will cause the carpet to buckle.

n) Perimeter carpet shall be accurately cut and tightly fitted against walls. Releasable type adhesive shall be applied in strips beneath all perimeter carpet tiles. Where the edge of the carpet tile floor adjoins another flooring, a reducing strip shall be installed.

o) Apply carpet and adhesive in accordance with manufacturers' instructions.

p) Verify carpet match before cutting to ensure minimal variation between dye lots.

q) Double cut carpet, to allow intended seam and pattern match. Make cuts straight, true, and un-frayed.

r) Locate seams in area of least traffic.

s) Join seams by hot adhesive tape method. Form seams straight, not overlapped or peaked, and free of gaps.

6.7.3.2 Clean-up & Protection

a) After the installation of the carpet is complete, all packaging materials, carpet waste, and debris shall be removed, and all carpet surfaces shall be thoroughly vacuumed. All carpet that has been damaged, improperly installed, or soiled beyond cleaning shall be removed and replaced.

b) Remove excess adhesive and other surface blemishes using cleaner recommended by carpet tile manufacturer. Replace carpet tile where blemishes cannot be removed. Trim protruding face yarn from carpet tile surface. Vacuum carpet tile using commercial quality vacuum cleaner with face-beater element, as per manufacturer's instruction.

c) Protect installation with a heavy, non-staining building paper. Do not use a moisture barrier such as plastic film. Do not permit traffic or installation of furniture on adhered areas for at least 24 hours. Maintain protection intact until construction time of handing over / commissioning commences.

d) Include manufacturer's maintenance procedures, recommended maintenance materials, and suggested schedule for cleaning.

6.8 Painting / Coatings

6.8.1 General

6.8.1.1 Scope of Work

a) This section covers the materials, labor, equipment, tools, accessories, workmanship, etc. required for the completion of all works related to painting of interior and exterior surfaces of buildings that are indicated on the drawings or as noted in finishes schedule / specifications. Workmanship shall be in accordance with BS 8000, Part 12.

b) All necessary preparation shall be carried out, including but not limited to cleaning surfaces, caulking joints, puttying of defects and nail holes, masking, prime all new / unpainted surfaces and materials, paint all gypsum wallboard wall and ceiling surfaces, doors and jambs, window jambs

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and trim, door casing, base board, shelving and other architectural trim elements and protecting of non-painted surfaces / materials.

c) CONTRACTOR shall provide all labour, materials, scaffolding, tools and implements required for preparation of surfaces, application of paint, making good any defects and cleaning of any paint splashes to other work.

d) All materials shall be of such quality as to produce first class and durable finishes and to be at least equal to the requirements of the relevant Project Standards. Full details of proposed materials and manufactures shall be submitted to the COMPANY for approval. Failure to do so may lead to rejection of materials on site and failure to meet programme requirements.

e) Unless otherwise approved by the COMPANY, all the different paints forming part of the system applied to a particular surface shall be supplied by the same manufacturer. Primers used for coating the surfaces of the pre-primed components shall be exempt from this requirement.

f) The SUB-CONTRACTOR shall not change the source of supply of any paint, or the formulation without first obtaining the approval of the COMPANY.

g) Paints shall have the correct consistency to suit the method of application. It shall normally be supplied with a consistency suitable for brush application and the SUB- CONTRACTOR shall include in his proposal, information relating to the consistency of undercoat, gloss and similar paints which are to be supplied by the MANUFACTURER for spray application.

h) Without restricting the extent of the work to be performed, the work shall include, but is not necessarily limited to, the following.

   i. Ferrous metals: Finish painting of exposed surfaces of ferrous metal work, including galvanized both exterior and interior, including metal fabrications, metal doors, frames, handrails and ladders

   ii. Masonry / Concrete: Painting on plastered concrete block, Ceiling, Concrete walls/parapets, boundary walls, etc. In case of external application, coatings shall be UV- resistant

   iii. Wood: finishing of all exposed wood, both exterior and interior

   iv. Gypsum board / cement fibre board: painting of all exposed Gypsum board/ cement fibre board

   v. Mechanical and electrical equipment: painting of ducts, piping, conduit, equipment, etc. except for items having acceptable factory finish

   vi. Concrete and Asphalt Traffic Surfaces: all striping required for parking stalls, walkways, carriageway, Curb Marking, Zone Marking, etc.

i) The following items shall be painted / coated as specified elsewhere in the respective specifications.

   i. Structural steel according to AGES-SP-01-002 Structural Steel Supply, Fabrication and Erection Specification.

   ii. Metal fabrications, Metal joists according to COMPANY Specification for Architectural Specialties, Metal, Wood and Plastics.

   iii. Roof deck and Roof hatch according to COMPANY Specification for Prefabricated Buildings

   iv. Metal doors, frames and Rolling counter fire doors according to COMPANY Specification for Doors and Windows

   v. Any other material in these Specifications noted to be pre-finished

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j) The following items shall not be painted under this section of the specification:

i. Pre-engineered steel building structure, roof panels, and siding

ii. Pre-finished roofing and siding

iii. Structural steel

iv. Metal HVAC roof curbs

v. Stainless steel items

vi. Pre-finished wood panelling

vii. Aluminum items with anodized finishes

viii. Toilet partitions, urinal screens, and shower compartments

ix. Acoustical materials

x. Overhead door

xi. Any other material in these Specifications noted to be pre-finished

6.8.1.2 Related Work Specified Elsewhere:

a) See AGES-SP-01-017 Masonry Wall and Thermal & Moisture Protection.

b) See Plastering, Section 6.2 herein.

c) See Gypsum Drywall, Section 6.3 herein.

6.8.1.3 Submittals

a) Products and installation shall be in accordance with the requirements of Architectural Design Basis AGES-SP-01-013. The CONTRACTOR shall submit a complete list of all paints intended to be used for each specific area to the COMPANY for approval before applying and it shall include the following:

i. Manufacturer's data sheets on each product to be used including label analysis, instructions for handling, storing, and applying each coating material, the submittal shall include:

1) Material List: An inclusive list of required coating materials. Indicate each material and cross-reference specific coating, finish system, and its application

2) Submit Block fillers, primers, and finish-coat materials that are compatible with one another and with the substrates indicated under conditions of service and application, as demonstrated by manufacturer based on testing and field experience

b) Mock-Up: Provide a mock-up for evaluation of surface preparation techniques, application workmanship, colors and finishes for all substrates. Before painting with a particular system commences, the CONTRACTOR shall paint test areas or sample panels to demonstrate that the specified thickness and finish to the paint film is being obtained. The paints, equipment and method of application used for test areas or sample panels shall be representative of those to be used for the actual work. Approved test areas or sample panels shall be retained and shall form the standard for all subsequent work.

c) A complete display of color samples and details for color selection process.

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d) VOC (Volatile Organic Compound) Classification: CONTRACTOR to submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013.

6.8.1.4 Handling and Storage

a) All painting materials shall be delivered to the site in the manufacturer's original sealed containers with all labels intact and legible. Acceptable project materials and related equipment shall be stored at the project site in a suitable location restricted to these materials. Practices shall comply with all health and fire regulations.

b) Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

c) Furnish extra 1 gallon paint materials from the same production run as the materials applied. Package with protective covering for storage and identify with labels describing contents.

6.8.2 Materials

6.8.2.1 Products

Provide block fillers, primers, rust-inhibiting rimer pigmented sealer, finish coat materials and related materials that are compatible with one another and with the substrates indicated under conditions of service and application, as demonstrated by the MANUFACTURER, based on testing and field experience.

6.8.2.2 Material Quality

a) A paint is composed of pigments, solvents, resins, and various additives. The pigments give the paint color; solvents make it easier to apply; resins help it dry; and additives serve as everything from fillers to anti fungicidal agents.

b) Provided painting and finishing products for use in the work shall be of best quality for each particular kind of material required. Primers and undercoat paint produced by the same manufacturer of the finish coats shall be used. Paint material containers not displaying manufacturer's product identification are not acceptable.

c) All paint and coating systems shall be in accordance with Architectural Design Basis AGES-SP-01-013 as well as its limits on Volatile Organic Compounds (VOC).

6.8.2.3 Colors

As selected by COMPANY.

6.8.2.4 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

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6.8.2.5 Minimum Coating Thickness

Applied coating thickness for primer, intermediate, barrier and finish coats shall be measured as dry film thickness (DFT) unless otherwise specified. Provide a total dry film thickness of the entire system as recommended by the manufacturer and complying with minimum coating thicknesses specified for each layer in this specification.

6.8.2.6 Selection of Paint System

a) Each type and grade of coating / painting system has varying application and performance properties. Different substrates require different types of primer / sealer. Special considerations shall be given to determining the existing condition of surfaces to be coated as well as in selecting the appropriate coating / painting system.

b) Selection of coating / painting system shall be based on high performance level, durability period, weather resistance, algal resistance, UV resistance, alkali resistance, scrub resistances, chemical resistance, waterproofing characteristic, etc. Application of the system shall be as per manufacturer's recommendations.

c) Coating materials and equipment shall be designed for the surface to be coated: finish coats shall be compatible with prime coats; prime coats shall be compatible with surface to be coated. Tools and equipment shall be compatible with coatings to be applied.

d) Colors shall be mixed by MANUFACTURER to match submittals reviewed by the COMPANY.

e) Vary tints of succeeding coats to permit identification.

f) Coating / painting system systems for some common substrata are given below.

6.8.2.7 Floors

a) Epoxy based coatings for concrete flooring of low traffic areas (e.g. Service areas, Mechanical rooms, Storage areas, etc.) shall be as follows:

   Solvent free, high build, epoxy resin floor coating with 2 coats (first coat and topcoat) of 200 microns DFT each. If anti-slip texture is used, DFT of the topcoat shall be 250 microns.

b) Coating for concrete flooring of high traffic areas / industrial activities (e.g. Workshops, Warehouses, etc.) shall be as follows:

   i. Heavy duty & Self-levelling epoxy resin based floor topping consisting of graded aggregates bound in a pigmented epoxy resin binder, to provide dense, impervious and chemical resistant flooring, 5 mm thick.

   ii. Epoxy floor coating shall consist of the following minimum properties:

1)	Compressive strength:	55N/mm2	BS 6319 Pt 2
2)	Flexural strength:	27N/mm2	BS 6319 Pt 3
3)	Absorption:	0.06%	ASTM C413

c) Inspect areas and conditions to receive flooring prior to installation. Do not start installation until unsatisfactory conditions are corrected. Installation of flooring shall be considered an acceptance of the substrate.

d) Preparation of Epoxy Subfloors:

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i. Do not install the flooring until the concrete subfloor has thoroughly cured. Do not install flooring sooner than 28 days after concrete has been cast. There shall be no exception to this rule. Perform required tests to determine that the concrete has cured

ii. The concrete subfloor scheduled to receive epoxy flooring shall be prepared by any of the following methods:

1) Vacuum shot blasting

2) Grit blasting

3) Acid Etching

iii. The subfloor shall be free of hardening compounds, dust, scale, and other deleterious matter immediately prior to installation of flooring, broom cleaned, and then vacuum cleaned. If acid etching has been employed the floor shall be rinsed completely with clean water and allowed to dry

iv. The prepared surface shall be primed by a primer recommended by the manufacturer of the epoxy material. The epoxy flooring shall be applied after a minimum of 12 hours after application of the primer or as recommend by the manufacturer

e) Application of Epoxy:

i. Install the epoxy flooring with non-slip aggregate per the MANUFACTURER'S written instruction.

ii. The coating should be 3-5 mm thick including the slip resistant application

f) Coating for concrete pavement with high vehicular traffic (e.g. car Parking areas and Garages), shall be as follows:

i. Multilayer modified polyurethane based flooring system - skid resistant, flexible and waterproof. Three layers of the system consist of primer (200 microns WFT), membrane (750 microns WFT) and wear course (140 microns WFT) with anti-slip grains.

6.8.2.8 Internal Wall surfaces

a) Internal coating / painting system for plastered masonry (concrete block) surfaces, plastered concrete surfaces and gypsum board / cement fibre board surfaces shall be as follows:

i. Primer Coat – One coat (min. 35 micron DFT) of High quality emulsion primer based on alkali resistant binder.

ii. Intermediate Coat – Two coats (min. 130 micron DFT each) of acrylic copolymer based putty.

iii. Finish Coat – Two coats (min. 75 micron DFT each) of high quality acrylic polymer based emulsion paint with good flow and smooth finish.

6.8.2.9 External Wall surfaces

a) External coating / painting system for plastered masonry (concrete block) surfaces, plastered concrete surfaces and cement fiber board surfaces shall be as follows:

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i. Primer Coat – One coat (min. 150 micron DFT) of High build epoxy coating based on epoxy resin of high molecular weight.

ii. Intermediate Coat – One coat of epoxy product formulated to be applied by spray equipment to give a textured surface.

iii. Finish Coat – Two coats (min. 40 micron DFT each) of polyurethane topcoat having excellent resistance against weather / ultraviolet light.

6.8.2.10 Soffits of slabs and beams

All beam surfaces and soffits of all slabs above the false ceiling shall be painted with light colored, approved concrete protective coating.

6.8.2.11 Un-plastered External Concrete Surfaces

All un-plastered external concrete surfaces shall be painted with a concrete protective painting system approved by COMPANY.

6.8.3 Execution

6.8.3.1 Job Conditions

a) Comply with manufacturer's recommendations as to environmental conditions under which paints and coating systems can be applied.

b) Do not apply in areas where dust is being generated.

c) Cover or otherwise protect finished work of other trades and surfaces not being painted concurrently or not to be painted. The CONTRACTOR shall propose a finished schedule to COMPANY for approval before construction.

d) Take all protective measures required to prevent accident and fire, to safeguard equipment and personnel, and to prevent interference with normal painting operations.

e) Store materials and equipment, mix paints, and clean tools and equipment in areas assigned for this purpose. Maintain these areas in clean and orderly condition taking precaution to prevent fires. Keep paint at room temperature.

f) In general, perform painting after other work has been substantially completed. When so directed, paint at times and in locations which will best expedite progress and completion of the project.

6.8.3.2 Surface Preparation

a) Clean and prepare surfaces to be painted in accordance with the manufacturer's instructions for each particular substrate condition.

b) Remove hardware and hardware accessories, plates, machined surfaces, lighting fixtures, and similar items already installed that are not to be painted. If removal is impractical or impossible because of size or weight of the item, provide surface-applied protection before surface preparation and painting. After completing painting operations in each space or area, reinstall items removed using workers skilled in the trades involved.

c) Cementations surfaces: Prepare concrete, concrete masonry block, and plaster to be painted. Remove efflorescence, chalk, dust, dirt, grease, oils, and release agents. Roughen as required to

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remove glaze. If hardeners or sealers have been used to improve curing, use mechanical methods of surface preparation.

d) Clean concrete floors to be flush with clean water to remove acid, neutralize with ammonia, and rinse; allow drying and vacuuming before painting.

e) Plaster Surfaces: All defective plastering is to be cut out, undercut edges trimmed and made good to the satisfaction of the COMPANY before decoration is commenced. All holes in internal plaster faces are to be made good with 'Alabastine' or other approved hard stopping, the edges being closely knit and the filling rubbed down level with the surrounding surface.

f) Building Board Surfaces: All building board surfaces to be distempered or painted are to have all nails painted with aluminium paint and all nail holes, etc., stopped with 'Alabastine' or other approved hard stopping which is to be rubbed down flush.

g) Wood Materials: Clean surfaces of dirt, oil, and other foreign substances with scrapers, mineral spirits, and sandpaper. Sand smooth surfaces exposed to view and dust off. Prime, stain, or seal wood to be painted immediately upon delivery.

h) All knots and resinous parts in wood surfaces to be painted are to be treated with two thin coats of knotting well brushed in. All cracks, crevices and holes in wood surfaces to be painted are to be scraped out, treated with one coat of thin primer and stopped with 'Alabastine'. New surfaces shall be well rubbed down and brushed clean, all knots shall be given two coats of best shellac knotting, an approved wood primer shall then be applied followed by all holes and imperfections being stopped or filled with an approved type filler, and a further coat ‘of primer applied. The specified number of undercoats shall then be applied followed by the finishing coat.

i) All ironmongery and associated equipment shall be removed prior to commencement of preparatory processes cleaning etc. and shall be re-fixed upon completion of the final coat.

j) Ferrous Metals: Clean non-galvanized ferrous-metal surfaces that have not been shop-coated; remove oil, grease, dirt, loose mill scale, and other foreign substances. Use solvent or mechanical cleaning methods that comply with recommendations of the Steel Structures Painting Council. Touch up with the same primer as the shop-applied prime coats that have been damaged. All exposed metal surfaces shall be protected against corrosion as per COMPANY specification for painting, AGES-SP-07-004 Painting and Coating.

k) New metal work to be painted shall be thoroughly cleaned and all traces of dirt, grease and other unwanted adhering matter removed. Any rust or mill scale etc. shall be removed with a powered wire brush or chipping hammer and rubbed down. The metalwork shall then be primed with an approved metal primer before the application of the undercoat and finishing coats. Where surfaces have been primed at the Manufacturer's Works all damaged portions shall be thoroughly cleaned prepared and touched up with an adequate coating of primer overlapping the adjacent sound work by at least 25 mm.

l) Galvanized Surfaces: Galvanised surfaces to be painted shall be as per COMPANY specification for painting, AGES-SP-07-004 Painting and Coating. Clean galvanized surfaces with non-petroleum-based solvents so that the surface is free of oil and surface contaminants. Remove pre-treatment from galvanized sheet metal fabricated from coil stock by mechanical methods.

m) Gypsum Board: Surfaces shall be dry and shall have all loose dirt and dust removed by brushing with a soft brush, rubbing with a dry cloth, or vacuum cleaning prior to application of first-coat material. Repair blemishes, irregularities and damaged surfaces.

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n) Material Preparation: Carefully mix and prepare paint materials in accordance with the manufacturer's directions.

o) Concrete and asphalt Traffic surfaces: The walkway, road surface shall be clean, dry and free from any loose detritus, mud and old flaking road marking materials or in accordance with the manufacturer's recommendations prior to application.

p) Surface Preparation Materials: Stopping for plaster shall be proprietary semi-hydrate I cellulose filler and shall be to the approval of the COMPANY. Knotting for wood surfaces shall comply with BS 1336. Stopping for wood shall be a proprietary general purpose wood and metal stopping and shall be to the approval of the COMPANY. Where the COMPANY permits the use of fillers for filling large holes in wood, a solvent less epoxy putty shall be used.

6.8.3.3 Application

a) Application shall be in accordance with the recommendations for good practice per BS 6150 and BS 8000, Part 12.

b) Mix and prepare paint materials and apply coatings by brush, roller, spray, or other applicators according to manufacturer's written instructions.

c) Use skilled painters under competent supervision to apply material in strict accordance with manufacturer's directions, including coverage rates and film thicknesses. Apply each coat in such a manner as to produce an even film of uniform thickness, free of runs, sags, brush marks, visible paint laps, and skipped or missed area. Completely coat crevices and corners but avoid clogging lines or leaving undue amount of paint.

d) Comply with recommendations of the paint manufacturer for drying time between coats.

e) No thinners shall be used without the consent of COMPANY.

f) Prepared surfaces shall be brushed or cleaned and shall be dried immediately before paint is applied.

g) Painted surfaces to be over painted shall be cleaned before a further coat of paint is applied. Dust shall be removed by brushing. Oil, grease or other contaminating matter shall be removed by washing and swabbing with clean white spilt of proprietary water mixtures of solvent and detergent which shall be to the approval of the COMPANY. Painted surfaces which have become contaminated by salt shall be washed with clean water.

h) The method of application shall be selected to ensure that the paint is applied in a uniform manner to the prescribed film thickness without any runs, sags or other blemishes, Compressed air used for spray application shall be adequate in pressure and volume to meet the work requirement and be free from oil and water contamination. Traps, separators and filters shall be emptied and cleaned regularly.

i) All priming paints shall be applied by brush except for etch primers which may be applied by brush or spray and holding primers which shall normally only be applied by spray.

j) Undercoats shall be applied evenly to the surface and if applied by brush, the paint on each section of the work shall be brushed alternately in directions at right angles using firm pressure on the brush before laying off using lighter finishing strokes. Laying off on timber surfaces shall be in the direction of the grain and towards the light on large surfaces of walls and ceilings.

k) Gloss and eggshell paints shall be applied evenly to the surface in a similar manner to that specified for undercoats. If application is by means of brushes, a wet edge shall be maintained while surfaces

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are being painted and the laying off shall ensure that there is no visible evidence of lapping in the finished paint film.

l) Unless otherwise approved by the COMPANY paint except for emulsion and masonry paint, shall not be applied by rollers.

m) Paints shall be applied as evenly as possible to provide a smooth coating of uniform thickness.

n) Intervals between coats shall comply with the paint MANUFACTURER recommendations, and generally shall be kept to the absolute minimum in order to prevent contamination between coats. Where contamination occurs between coats, this must be completely removed by washing with detergent solution and rinsing with clean water. The surface shall be allowed to dry before application of paint.

o) Damaged areas of priming coats or undercoats shall be made good before further coats of paint are applied. The types of paint and the number of coats used for making good shall be the same as those existing on the damaged surface. The edges of damaged areas of paint films on surfaces for which a superior decorative or decorative finish is specified shall be rubbed down to a feather edge and the coats of paint used for making good shall be brushed out at the edges to ensure than no lips are visible on the finished surface.

p) At each fire extinguisher shown on the plans, the SUBCONTRACTOR shall paint a background 300 mm wide and 1000 mm high. Color shall be Industrial Enamel B54R18, OSHA Red or BS color code equal.

q) Pipe and conduits shall be painted the color of the surface against which they are installed or seen.

r) All pavement, road, floor markings and striping shall be performed as shown and called for on the drawings. All painting and striping shall be with two coats of Traffic Marking Paint, type to COMPANY approval.

s) Where required by the Contract, the CONTRACTOR shall be responsible for the erection, maintenance and dismantling of all scaffolding. The scaffolding arrangement shall be in accordance with BS EN 12811-1 or BS 5974 as applicable and shall be subject to approval by COMPANY.

t) Scaffolding used for painting shall be retained in position whilst work is being carried out and for a period up to three days after completion, or alternatively, other means shall be provided to give reasonable access for the COMPANY to inspect prepared surfaces and paint films.

6.8.3.4 Cleaning, Disposal and Protection

a) Touch up and restore damaged finishes.

b) Remove spilled, splashed, or splattered paint from all surfaces. Keep surfaces clean and free from dust during coating and drying.

c) Remove surplus materials, containers, equipment, and tools promptly when not needed. Collect oily rags and waste daily; destroy them or deposit them in tightly covered metal containers. Clean up and dispose of debris at close of each workday and upon completion of all painting operations. All debris and surplus material shall be disposed off-project limits in a manner meeting all Local Authority requirements.

d) Provide ‘Wet Paint’ signs to protect newly painted finishes. After completing painting operations, remove temporary protective wrappings provided by others to protect their work.

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e) At the completion of the project leave the storage space clean and in the condition under which it was taken.

6.9 Wall Coverings

6.9.1 General

6.9.1.1 Scope of Work

a) This Section covers vinyl wall coverings to be furnished and installed primarily for its decorative effect, while other wall covering is also designed to achieve a high degree of serviceability. Vinyl wall covering shall be installed where indicated as per the finishes schedule.

b) Workmanship for the wall covering installation shall be in accordance with BS 8000 Part 12.

6.9.1.2 Related Work Specified Elsewhere:

a) See Resilient Flooring, Section 6.6 herein

b) See Plastering, Section 6.2 herein.

c) See Gypsum Drywall, Section 6.3 herein

d) See Painting / Coatings, Section 6.8 herein.

e) AGES-SP-01-013 Architectural Design Basis, see 'APPENDIX A - Typical Finishing Schedule'

6.9.1.3 Submittals

a) Samples of each color, measuring a minimum of 150 mm by 200 mm shall be submitted to the COMPANY. These samples shall be the basis of all vinyl wall covering installed.

b) Submit documentation that the manufacturer's wall covering meets Federal Specifications CCC-W-408D and CFFA W-101D for wall covering products.

c) The manufacturer shall certify that the vinyl wall covering supplied meets Federally specified limits for flame, smoke development and flash over according to the required ASTM F793 - 10a Standard Classification of Wall Covering by Use Characteristics and NFPA 286 tests.

d) VOC (Volatile Organic Compound) Classification: CONTRACTOR to submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013.

6.9.1.4 Delivery, Storage and Handling

a) All necessary materials for installation including wall covering, primer and adhesives must be delivered to the job site undamaged. All containers are to be clearly marked with the manufacturer's identification label. Store vinyl wall covering rolls and materials in a clean, dry area where temperature and humidity remain constant and within

b) Materials shall be stored in a clean, dry storage area with a minimum constant temperature of 5°C. Materials shall be stored flat and not in an upright position.

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6.9.2 Materials

a) Vinyl wall covering materials shall conform to the following standards and quality levels. Manufacturer and product shall be subject to submittal review and approval by the COMPANY.

Vinyl wall covering	Fed Spec CCC-W-408D, Type II, to COMPANY approval Wall covering shall contain mildew and germicidal additives. Final selection of colors and textures shall be made by the COMPANY and shall conform to CFFA-W-101-A, vinyl wall covering shall comply with ASTM E-84 for fire hazard classification
Adhesives	As recommended by the vinyl wall covering manufacturer for the surface
Primer	As recommended by the vinyl wall covering manufacturer

6.9.3 Execution

6.9.3.1 Installation

a) Unbacked flexible PVC (vinyl) sheet shall be to BS EN ISO 10581, minimum 1.2 mm thick to walls.

b) The wall covering material shall extend from floor to ceiling. Patterns shall match both horizontally and vertically. Before installation of wall covering, each wall shall receive a primer coat.

c) Areas to receive wall covering shall have a constant temperature of 15°C for at least 3 days prior to and all during installation. Plaster and masonry surfaces to receive wall covering finish shall not contain more than 5.5 percent moisture.

d) Panels shall be used in exact order as they are trimmed from the rolls. Areas over doors and other openings shall be filled in with panels cut in consecutive order from the rolls.

e) Fabric shall be smooth to surfaces with a stiff bristled brush or flexible broad-knife to eliminate air bubbles and ensure adhesion.

f) Where applicable, wall coverings shall be installed before the installation of plumbing, millwork, bases, and other accessories which are to be installed on or into the wall surface.

g) Apply adhesive in a thin film and spread evenly with a notched or plain trowel, then even out with a roller as recommended by the manufacturer.

h) Lay wall tiles or wall sheeting strictly in accordance with the manufacturer's instructions. If these instructions call for heating the tiles prior to laying or sheets shall be laid immediately after the adhesive is applied.

i) Vinyl wall covering shall be installed with no horizontal joints, and with no vertical joints within 150 mm of outside corners. Applicable procedures shall include stretching, rolling, and otherwise smoothing to ensure even surfaces and invisible joints.

j) After laying, thoroughly clean the tiling or sheeting with water and a detergent recommended by the manufacturer. Apply two coats of matt emulsion polish.

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6.9.3.2 Spare Materials

Spare material equal to 1.5 percent of vinyl wall materials in each type / color shall be furnished for use in future maintenance work.

6.9.3.3 Cleaning

Upon completion of the work, all surplus materials, rubbish, and other debris resulting from installation of the wall covering shall be removed.

6.10 Wall Skirtings

6.10.1 General

6.10.1.1 Scope of Work

a) This section covers the furnishing of materials and installation for glazed ceramic tile wall skirting, marble tile wall skirting, hard wood skirting, Terrazzo tile skirting and PVC skirting tiles.

b) The skirting work shall be coordinated with the floor finishing works and electrical / instrument cabling works, as required.

c) Skirtings shall be securely fixed to the partition framing by approved fixing materials and method. Skirting tiles shall be installed where indicated as per the finishes schedule but as a minimum below mentioned to be followed:

   i. 100 mm high glazed ceramic tile skirting shall be provided to areas with ceramic tile flooring (width to match the floor Tile size)

   ii. 100 mm high marble tile skirting shall be provided to areas with marble tile flooring (width to match the floor Tile size)

   iii. 100 mm high hard wood skirting boards with paint finish shall be provided to areas with carpet finish

   iv. 100 mm high PVC sit-on skirting shall be provided to vinyl floor areas

   v. 100 mm high Terrazzo skirting tile (width to match the floor Tile size)

   vi. 100 mm high Chemical Resistant skirting tile (width to match the floor Tile size)

   vii. The color and pattern / shape of skirting tiles / boards shall match with the floor finishes and shall be approved by COMPANY

6.10.1.2 Related work specified elsewhere

a) See Tile, Section 6.4 herein

b) See Resilient Flooring, Section 6.6 herein

c) See Carpeting, Section 6.7 herein

d) AGES-SP-01-013 Architectural Design Basis, see ‘APPENDIX A - Typical Finishing Schedule’

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6.10.1.3 Submittals

a) The color and pattern / shape of skirting tiles / boards should match with the floor finishes and shall be as specified and approved by the COMPANY.

b) Products and installation shall be in accordance with the requirements of Architectural Design Basis AGES-SP-01-013. Submit manufacturer's product information, catalogues, specifications, installation details and drawings, product samples and other data as may be required to certify compliance with performance requirements.

c) Skirting tile or board samples shall be submitted to the COMPANY after colors and pattern have been selected. The samples shall include trims and special shapes. At least two samples shall be submitted in each color selected.

d) VOC (Volatile Organic Compound) Classification: CONTRACTOR to submit materials, including primers, undercoats, sealants, adhesive and finish-coat materials that meet local air quality management regulations as per Architectural Design Basis AGES-SP-01-013.

6.10.1.4 Handling and Storage

All materials shall be packaged at the factory and shipped to the project site in unbroken grade-sealed containers. Material containers shall be unopened and shall be clearly marked with the manufacturer's name, brand and color coded grade seal. Materials shall be carefully handled and stored in weatherproof enclosures.

6.10.2 Materials

a) Wall skirting material shall conform to the following standards and quality levels. Alternate products will be considered upon submittal reviews and approval by the COMPANY.

b) Glazed ceramic tile: Shall conform to ANSI A137.1, glazed tile with cushioned edges and trim edged with lead free finish. Minimum thickness shall be 6 mm. Type to COMPANY approval.

c) Marble Skirting: Shall conform to ASTM C503, Group A, sound, dense, homogeneous and free from defects. Minimum thickness shall be 15 mm. Type to COMPANY approval.

d) Granite Skirting: Granite shall comply with ASTM C615 / C615M. Granite shall be standard grade, sound, dense, homogeneous and free from defects. Minimum thickness shall be 15 mm. Type to COMPANY approval.

e) Hard Wooden Skirting: Skirting shall be constructed of a durable rigid material such as painted wood, exterior grade wood composites, exterior grade plywood. Minimum 15 mm thick, paint finished, moisture-proof. Type to COMPANY approval.

f) PVC Skirting: Fireproof, moisture proof, and wear proof, germ repellent and environmentally friendly. Thickness of board shall be as per manufacturer standards.

g) Terrazzo Skirting tile: The material shall be homogenous, dense and compact structure, each color, grade, type, and variety of granular materials from shall be from single source with resources to provide materials of consistent quality in appearance and physical properties. Tiles shall be 10 mm rectified on all four sides and edges seamed (chamfered).

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6.10.3 Execution

6.10.3.1 Installation

i. The installation of glazed ceramic / marble skirting shall be as manufacturer's written instruction and as per section 6.4 herein.

ii. The surface preparation and installation of Wooden and PVC skirting boards shall be strictly as per the manufacturer's written instructions and installation drawings.

6.10.3.2 Surface Preparation

a) The joints shall be raked out to a depth of at least 10 mm in masonry walls.

b) In case of concrete walls, the surface shall be hacked and roughened with wire brushes. The surface shall be cleaned thoroughly, washed with water and kept wet before skirting is commenced.

6.10.3.3 Spare Materials

Spare material equal to 2.5 percent of wall skirting materials in each type / color shall be furnished for use in future maintenance work.

6.10.3.4 Cleaning

Upon completion of the work, all surplus materials, rubbish, and other debris resulting from installation of the wall skirting shall be removed.

6.11 Stone Cladding / Lining Panels

6.11.1 General

6.11.1.1 Scope of Work

a) This section covers the furnishing of materials and installation of Stone Wall Cladding / Lining Panel Systems. The stone wall cladding system may be considered with specific COMPANY approval to provide an attractive finished exterior appearance to the building walls.

b) The work shall be coordinated with all adjacent finishing and other works including electrical / cabling / lighting works as required.

c) The system shall use natural stone or manufactured artificial stone panels bonded to lightweight backing / support structure by manufacturer, complete with stone facings and reinforcement as required. Connection and anchorage hardware, interlocking channels, anchor plates, Z-sections, angle clips and threaded inserts shall be provided to detailed design requirements and manufacturer's recommendation. Sealant materials shall be tested for compatibility with the natural stone facings specified.

d) All products and design proposals shall be agreed with COMPANY.

6.11.1.2 Related work specified elsewhere

a) AGES-SP-01-013 Architectural Design Basis

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6.11.1.3 Submittals

a) Submit manufacturer's product information, catalogues, specifications, installation details and drawings, product samples and other data as may be required to certify compliance with performance requirements.

b) Submit independent laboratory certified test reports showing compliance with specified performance characteristics.

c) The coloration and pattern / texture of panels shall be approved by COMPANY, including polished, honed, sandblasted, flamed or tooled finishes.

d) Structural calculations comprising structural adequacy and stability of the proposed cladding system, element design for stone, aluminium or metal cladding and deflection check on major load carrying members, as applicable.

6.11.1.4 Handling and Storage

All materials shall be packaged at the factory and shipped to the project site in unbroken grade-sealed containers. Material containers shall be unopened and shall be clearly marked with the manufacturer's name, brand and seal. Materials shall be carefully handled and stored in weatherproof enclosures. Store materials protected from exposure to harmful weather conditions. Handle material and components to avoid damage. Protect material against damage from elements, construction activities, and other hazards before, during and after installation.

6.11.2 Materials

CONTRACTOR shall submit materials and system information to conform with BS 8298, and / or ASTM C1242, ASTM C1354, and ASTM C1670 / C1670M or equivalent - to be agreed with COMPANY.

6.11.3 Execution

Verify substrate conditions are acceptable for product installation in accordance with manufacturer's instructions. Verify modules are sized to receive stone wall panel system in accordance with manufacturer's acceptable tolerances. Verify actual measurements / openings by field measurements before fabrication. Confirm recorded measurements on shop drawings.

6.11.3.1 Installation

Install stone wall panel system plumb, level, and true to line with manufacturer's prescribed tolerances and installation instructions. Provide supports and anchor in place. Provide separation of aluminum materials from sources of corrosion or electrolytic action contact points.

6.11.3.2 Spare Materials

Spare material equal to 2.5 percent of materials in each type / color shall be furnished for use in future maintenance work.

6.11.3.3 Protection and Cleaning

Protect installed product's finish surfaces from damage during construction. Protect stone facing from damage from harmful contaminants. Repair or replace damaged installed products. Clean installed

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products in accordance with manufacturer's instructions prior to owner's acceptance. Remove construction debris from project site and legally dispose of debris.

6.12 Laboratory Furniture

Laboratory Furniture shall be provided as described and listed in Contract / Project specific documents.

6.13 Shop Furniture

6.13.1 General

6.13.1.1 Scope of Work

Under this Section, the CONTRACTOR shall furnish all labour, equipment, and materials for the design, construction and arrangement of shop furniture, seating and accessories that shall be furnished and installed in the warehouse / maintenance workshops and other buildings as applicable for the PROJECT.

6.13.1.2 Submittals

Provide product information, installation instructions and furniture layout drawings for all areas for COMPANY review and acceptance. The layout drawings shall identify all furniture with selected options and accessories. No work shall commence without COMPANY approval.

6.13.1.3 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

6.13.2 Materials

The hereunder requirements shall be updated by CONTRACTOR to Project codes / standards to be chosen together with COMPANY.

6.13.2.1 Sheet Metal

All sheet metal used in cabinet, drawer, shelf, top door, and base construction shall be heavy gauge, prime quality, leveled and re-squared, and free from any defects such as rust, scratches and ragged edges. All cabinets and components shall have wear-resistant baked-on enamel finish resistant to solvents.

6.13.2.2 Workbench Tops

All base cabinet countertops shall be satin finish, 45 mm thick hard maple with attachment hardware supplied.

6.13.2.3 Base Cabinets

a) Each base cabinet shall be constructed as a complete integral case with full sides, top and back. Sides shall have four punched holes, one near each corner of the side walls, for bolting cabinets

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together. Tops shall have knockouts for securing worktops or stacking cabinets. All fastener hardware shall be supplied.

b) Each cabinet shall be provided with a lock with two keys, with each group of cabinets shall have a master key.

6.13.2.4 Drawers

All drawers shall have 100% full extension and a one-drawer-at-a-time safety system to prevent tipping. Handles shall be continuous across the front of the drawer face and shall not protrude past the front of the cabinet when fully closed.

6.13.2.5 Cabinet Doors

All cabinet doors shall be provided with a lock with two keys, as well as being keyed to the group master key.

6.13.2.6 Base

All base cabinets shall be provided with a fork truck or pallet truck base to permit moving of fully loaded cabinets. Bases shall be pre-punched to allow stacking and shall have snap-on front and rear covers. Floor anchors and bolting hardware shall be supplied.

6.13.2.7 Accessories

A list of all drawer partitioning sets, labeling systems and other options shall be provided to the COMPANY for review and approval.

6.13.2.8 Industrial Seating and Stools

Heavy-duty metal construction seating or stools shall be provided at workbench stations that require tasks of extended duration. All seats and stools shall have steel glides.

6.13.3 Execution

6.13.3.1 Installation

a) Shop furniture shall be installed as shown on the shop drawings and as required by MANUFACTURER'S written installation instructions.

b) Upon completion of installation and adjustments, all labels shall be removed and all surfaces shall be cleaned as recommended by the MANUFACTURER.

c) All cartons, packing materials and rubbish due to shop furniture work shall be removed.

6.14 Miscellaneous Furniture, Furnishings and Equipment

6.14.1 General

6.14.1.2 Scope of Work

Under this section the CONTRACTOR shall furnish and install all miscellaneous equipment, including horizontal louver blinds, vertical blinds and floor mats.

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6.14.1.3 Related Work Specified Elsewhere

a) See Company Specification for Architectural Specialties, Metal, Wood and Plastics.

b) See Section 6.10 herein.

6.14.1.4 Submittals

Submit product data, installation and operation drawings and instructions, samples (as applicable) and manufacturer standard finishes and colors for each item for selection and approval by the COMPANY. No work shall commence without the approval of the COMPANY.

6.14.2 Materials

6.14.2.1 Horizontal Louver Blinds

a) Each horizontal louver blind unit shall consist of slats (available in PVC, aluminum, fabric and wood, in various finishes), headrail, tilter, tilt wand, tilt rod, cord lock, bottom rail, and locks, ladders, lift cords, tilt rod supports, installation brackets and valance.

b) Lifting and tilting mechanisms shall be constructed of noncorrosive, self-lubricating materials.

c) Units shall be fabricated in sizes to fill window and other openings by measurements taken in the field at the temperature of 23°C. For units to be installed between (inside) jambs, size shall be overall opening width and length less manufacturer's required tolerances. For units to be installed outside jambs, size shall be width and length as indicated on the drawings or in the Window Schedule, with terminations between units of end-to-end installations at center lines of mullion or other defined vertical separations between openings, less any manufacturer's required tolerances.

d) Support brackets shall be fabricated from metal, noncorrosive to unit hardware and adjoining construction and shall support blind units under conditions of normal use.

e) Slat materials, finishes, slat sizes and colors shall be as selected from the manufacturer's standard colors by the COMPANY.

f) Acceptable manufacturers shall be approved by COMPANY upon review of submitted product information.

6.14.2.2 Vertical Blinds

a) Each vertical blind unit shall consist of vanes (available in PVC, aluminum and fabric with various finishes), headrails, mounting brackets, tilt rod, carriers, vane hooks, traversing control, cord weight, valances, and 180° rotation ability of the vanes.

b) Traversing mechanisms shall be constructed of noncorrosive, self-lubricating material, allowing for smooth traversing with a minimum draw force.

c) Units shall be fabricated in sizes to fill windows and other openings by measurements taken in the field at the temperature of 23°C. For units to be installed between (inside) jambs, size shall be overall opening width and length less manufacturer's required tolerances. For units to be installed outside jambs, size shall be width and length as indicated on the drawings or in the Window Schedule, with terminations between units of end-to-end installations at center lines of mullion or other defined vertical separations between openings, less any manufacturer's required tolerances.

d) Support bracelets shall be fabricated from metal, noncorrosive to unit hardware and adjoining construction and shall support blind units under conditions of normal use.

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e) Vane materials, vane sizes, finishes and colors shall be selected from the manufacturer's standard colors by the COMPANY.

f) Products by manufacturers shall be considered for approval upon review of information submitted to the COMPANY.

6.14.2.3 Recessed, Roll-Up Floor Mat

a) Recessed, roll-up floor mats shall consist of tread rails (constructed of aluminum or vinyl), continuous hinge connectors (constructed of finished aluminum), tread surface (available in carpet, abrasive and vinyl) and continuous finished aluminum frame (to accommodate recessed cast-in or block-out installation).

b) Areas to receive mats shall be prepared as required with appropriate perimeter framing to accommodate adjoining materials.

c) Mats shall be fabricated in sizes as indicated on the drawings, less manufacturer's required tolerances.

d) Tread rail, hinges, inserts and perimeter framing finishes and colors shall be as selected by the COMPANY from manufacturer's standard colors.

e) Products by manufacturers shall be considered for approval upon review of information submitted to the COMPANY.

6.14.2.4 Deep Pit, Recessed Floor Grille

a) Deep pit, recessed floor grilles shall consist of grids (constructed of aluminum tread rails fabricated from 6063-T52 alloy, structurally joined to aluminum lock bars), tread surface (available in carpet, abrasive and vinyl), continuous finished aluminum frame (to accommodate recessed cast-in installation) and intermediate grid support framing (constructed of extruded aluminum matching the grid fabrication).

b) Floor grille pits shall function as a plenum connected to a remote dust extraction fan and filter system.

c) Areas to receive the grilles shall be prepared as required with appropriate perimeter framing to accommodate adjoining materials.

d) Grilles shall be fabricated in sizes as indicated on the drawings, less manufacturer's required tolerances.

e) Grid, lock bar, inserts and perimeter and intermediate framing finishes and colors shall be as selected by the COMPANY from manufacturer's standard color.

f) Products by manufacturers shall be considered for approval upon review of information submitted to the COMPANY.

6.14.3 Execution

6.14.3.1 Installation

a) Miscellaneous furniture, furnishings and equipment shall be installed as shown on the drawings and as required by MANUFACTURER'S installation instructions.

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b) Upon completion of installation and adjustments, all surfaces shall be cleaned as recommended by the MANUFACTURER.

c) All cartons, packing materials and rubbish due to miscellaneous furnishing's work shall be removed.

6.15 Movable Wall Partition System

6.15.1 General

6.15.1.1 Scope of Work

a) Under this section, the CONTRACTOR shall furnish all labour, equipment and materials required to erect unitized demountable full-height and partial-height metal partitions, including furniture components and accessories.

b) The movable wall / partition system shall offer maximum flexibility and reusability to accommodate frequent and quick relocation work without loss of materials, damage or modification to panels or to adjoining structures such as ceilings, fixed walls and floors.

c) The system must be unitized, non-progressive and modular, allowing the removal of individual panels from any location without disturbing adjoining units and providing interchangeability of panels and door units on the same module.

6.15.1.2 Related Work Specified Elsewhere

a) Architectural Design Basis - AGES-SP-01-013.

b) Electrical Engineering Design Guide - AGES-GL-02-001.

6.15.1.3 Submittals

Provide complete detailed shop drawings, erection drawings, specifications, and data covering all items being provided. drawings shall indicate elevations of each run of partitions, details of framing, location, and identification of each part, special details of construction, methods of assembly and enclosing, location and installation requirements of hardware, and finish. No work shall commence without the approval of the COMPANY.

6.15.1.4 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

6.15.2 Materials

6.15.2.1 Partition Requirements

a) Partition shall have a nominal thickness of 50 mm to 75 mm and feature a hanging system at each panel joint, including corners and finished ends. The hanging system shall have slotted standards capable of accommodating components of major landscape furniture systems, full height shelving and other wall mounted accessories such as tackboards, acoustical panels, chalkboards, etc.

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Design shall permit extension in two-way, three-way and four-way conditions without removing adjacent panels. Panel units of equal size, whether solid or glazed, including door units shall be overlapping to accommodate additional cabling for electrical and communication purposes.

b) The panel assembly shall have the capability for easy vertical adjustment of ± 38 mm for leveling of panel units to compensate for ceiling and floor irregularities.

6.15.2.2 Panels

a) Panels shall be fabricated with a minimum 18-gauge steel frame construction with panel faces of 20-gauge tension levelled steel with welded stiffeners or 26-gauge steel laminated to a honeycomb core. The factory assembled units shall be completely insulated with mineral wool and shall provide an overall sound transmission class (STC) of not less than 39 for solid, full-height panels. Panel faces shall be smooth and free from waviness, oil canning, or dimples and be capable of being field removable for easy repair if defaced.

b) Panel sizes shall be available in unit widths from 200 mm to 1500 mm, and a height up to 3000 mm.

c) Wall system panels shall meet the load bearing capacity standards for hanging component of 35 kg per linear 300 mm. Deflection shall not be 1/240 of the overall span under a load of 2.25 kg per 90,000 mm2.

d) Partition panel construction shall not exceed the following fire rating requirements when tested in accordance with the provisions of ASTM E84:

e) Flame Spread Rating: 5

f) Smoke Density Factor: 25

6.15.2.3 Glass and Glazing

Glass panels shall be furnished factory glazed and installed as indicated on the drawings. All glass shall be 6 mm thick tempered glass, unless indicated otherwise. All glass in glazed panel units shall be set in a continuous removable grey vinyl seal at factory. Panel frame construction, head and base assemblies shall be identical to solid panel construction.

6.15.2.4 Ceiling Track Assembly

Channel shall be constructed of 18-gauge steel and reinforced structurally to retain panels in a true vertical plane. The track shall be designed to accept a continuous sound and light seal gaskets and permit the use of Snap-on ceiling clips to promote mobility and eliminate penetration in the ceiling systems.

6.15.2.5 Wall Connection Assemblies

End fillers which joint partitions to fixed walls or columns shall be similar construction to the solid panel units. End filler panels shall fasten securely to adjacent panel units and be spring loaded to fit neatly into a channel abutting into the fixed wall or column without penetration. The wall channel shall be scribed and gasketed as required to provide a continuous light and sound seal.

6.15.2.6 Base Assembly

The track shall be constructed of a 16-gauge steel channel, with light and sound seal capabilities, and carpet adapters. Continuous base covers shall be provided which allow access for panel adjustment,

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installation of wiring and accommodation of outlet boxes without field cutting. All voids are to be insulated with mineral wool.

6.15.2.7 Doors and Frames

a) Doors shall be solid core with wood veneer, plastic laminate or painted faces as indicated in the Door Schedule. Doors shall be full height, 45 mm thick, and mortised to receive 1-1/2 pair of ball bearing butts and locksets.

b) Door frame shall be full height of 18-gauge steel with reinforced mortising to accept doors and shall have a continuous hollow vinyl tube molding stop to serve as a sound and light seal and act as a muting cushion. Frames shall be adjustable vertically for installation on varying floor conditions.

c) Openings specified for nominal 2100 mm high doors will be provided with a transom panel 45 mm thick, having the same construction of the doors, for installation in standard full height frame.

6.15.2.8 Door Hardware

a) All finish hardware, except cylinders and keys, shall be furnished by partition MANUFACTURER to simplify coordination and reinforcement and mortising of doors and frames. Lock cylinders and keys shall be furnished by the VENDOR furnishing hardware for doors in permanent building walls. Partition hardware shall be installed by the CONTRACTOR.

b) Finish hardware shall be in accordance with COMPANY Specification for Doors and Windows.

6.15.2.9 Finishes

a) All exposed surfaces shall be degreased, cleansed and treated with an etched phosphate rust resistant coating, followed by the application of a rust inhibitive primer coat, finished with a baked enamel paint finish.

b) Upholstered panels shall have a baked enamel finish prior to vinyl or fabric application. Paint colors, vinyls, fabrics and glass shall be selected by the COMPANY from the MANUFACTURER'S standard materials and colors.

6.15.2.10 Partition Accessories

a) Wall hung and free-standing partition accessories shall include storage units, drawer units, work surfaces, accent panels, coat hangers, tacks, filing units, lighting units, etc. shall be furnished by acceptable manufacturers and installed by the accessories MANUFACTURER or SUBCONTRACTOR approved by the accessories MANUFACTURER.

b) Adapter brackets for wall hung accessories shall be furnished by the partition MANUFACTURER.

c) A listing of wall hung and free-standing partition accessories shall be indicated in the Finish Schedule.

6.15.2.11 Electrical Requirements

Design of base and ceiling members shall permit extension of wiring at those locations. Fittings and flush plates shall be furnished by partition CONTRACTOR for locations of these items as shown on partition layouts only. Wiring and electrical installation work shall be in accordance with Electrical Engineering Design Guide AGES-GL-02-001.

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6.15.2.12 Stock Material

Additional partition material shall be furnished and delivered to the COMPANY for use in future partition relocations. Sizes, finishes and quantities shall be as indicated in the Finish Schedule. Material shall be placed in storage as directed by the COMPANY.

6.15.3 Execution

6.15.3.1 Installation

a) Partitions shall be installed after floor coverings, painting and suspended acoustical ceilings have been installed. Care shall be taken to avoid damaging installed work.

b) Partitions shall be installed rigid, level, plumb, flush, and in alignment, without cracks or loose parts.

c) A continuous and positive seal to prevent light and sound transmission shall be installed where partitions contact floors, ceilings, walls, and other surfaces.

d) Door and frame panels shall be installed without the use of fasteners into the floor.

e) Hardware shall be adjusted to proper operation.

f) Accessory items and attachments shall be installed by others at the locations indicated on the drawings and as directed by the CONTRACTOR. All wall-hung items shall be firmly secured in place.

g) Damaged or defaced surfaces of partitions or components shall be completely refinished with factory finish materials or shall be completely replaced as determined by the CONTRACTOR at no extra cost to the COMPANY.

6.15.3.2 Clean-up

All cartons, packing materials, and rubbish due to partition work shall be removed. Labels shall be removed from glass, and partition surfaces shall be cleaned as recommended by the partition MANUFACTURER. Partition accessories shall be cleaned and labels removed as recommended by the MANUFACTURER.

6.16 Furniture Requirements

6.16.1 General

6.16.1.1 Scope of Work

Under this section the CONTRACTOR shall furnish all labour, equipment and materials for the layout and arrangement of furniture and accessories that shall be furnished and installed in the Process and Non-Process Buildings as specified in Contract / Project specific documents. Refer definition of 'Process' and 'Non-Process Buildings' in Architectural Design Basis AGES-SP-01-013.

6.16.1.2 Related Work Specified Elsewhere

a) Laboratory Furniture to be provided as listed in Contract / Project specific documents

b) Shop Furniture, Section 6.13 herein.

c) Movable Wall / Partition System, Section 6.14 herein.

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6.16.1.3 Submittals

Provide product information, installation and assembly instructions, and furniture layout drawings for all indicated areas for COMPANY review and acceptance. The layout drawings shall identify all furniture with selected options and accessories. No work shall commence without COMPANY approval.

6.16.1.4 Manufacturers

a) The Product / material complying with the specification requirements shall be sourced from an Approved Manufacturer only as per COMPANY Approved Vendor List (AVL).

b) In case of non-availability of material / product from the AVL, CONTRACTOR can propose alternate material / product and shall follow the prequalification process to obtain COMPANY approval for the same.

6.16.2 Materials

The material specifications for furniture are as given below for guidance, unless specified otherwise in project specific drawings / documents.

6.16.2.1 The description for desks, returns and worktops is as follows:

ITEM NO.	DESCRIPTION
W1	Desk, finely crafted wood, 1800w x 900d x 725h, with modesty panel and side drawers
W2	Desk with return, wood veneer, 1800w x 900d x 725h, with modesty panel and side drawers.
W3	Desk, laminated finish, 1500w x 900d x 725h, with modesty panel.
W4	Work station, wood veneer, with work surface and 2 pedestals.
W5	Work station, laminated finish, with work surface and 2 pedestals.
W6	Reception desk, finely crafted wood, with high front panel.
W7	Engineering drafting station with telephone reference desk fitted with pedestal, 2 drawers and side table.
W8	Conference table, finely crafted wood, seating for 16.
W9	Conference table, wood veneer, seating for 14.
W10	Conference table, finely crafted wood, 1350 diameter x 700h.
W11	Conference table, wood veneer, 1200 diameter x 700h.
W12	Conference table, laminated finish, 1200 diameter x 700h.
W13	Roller sliding projection table.
W14	Printer table.
W15	Unit kitchen, type to COMPANY approval
W16	Shop furniture, Section 6.13 herein
W17	Conference table, finely crafted wood, 2000 diameter x 700h.

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W18	Built-in laminated countertop with knee space, drawers and cabinets (top of counter: 750 mm).
W19	Circular table with center support, laminated finish, 900 diameter x 700h.
W20	Training Table 600w x 1500d x 700h. Foldable, nestable training table with vertical & horizontal wire management, modesty panel. Tabletop shall be laminated with flat profile PVC edge banding and aluminium base with casters.
W21	Conference table, laminated finish, seating for 16 people
W22	Conference table, laminated finish, seating for 12 people
W23	1200w x 1200d x 700h table with single metal pedestal with glides. Plastic laminate top with vinyl edge.

6.16.2.2 The description for seating is as follows:

ITEM NO.	DESCRIPTION
S1	High-back chair, fully upholstered in genuine leather, “T” arms, pneumatic height adjustment, variable back stop.
S2	Mid-back chair, fully upholstered in genuine leather, “T” arms, pneumatic height adjustment, variable back stop.
S3	Mid-back meeting chair, fully upholstered in genuine leather, open arms, fixed base with glides.
S4	Professional’s chair, fabric, soft arm caps, pneumatic height adjustment.
S5	Professional’s chair, fabric, armless, pneumatic height adjustment.
S6	Standard back visitor’s chair, fabric, armless, steel base.
S7	Meeting chairs, fabric, open arms, fixed base with glides.
S8	Easy chair, fully upholstered in genuine leather, foam padded structure with reversible foam padded seat cushions and fiber filled back cushions, flame and fire resistant.
S9	3-seat sofa, fully upholstered in genuine leather, foam padded structure with reversible foam padded seat cushions and fiber filled back cushions, flame and fire resistant.
S10	Easy chair, fully upholstered in Dacron fabric, foam padded structure with reversible foam padded seat cushions and fiber filled back cushions, flame and fire resistant.
S11	3-seat sofa, fully upholstered in Dacron fabric, foam padded structure with reversible foam padded seat cushions and fiber filled back cushions, flame and fire resistant.
S12	Lecture chair, fully upholstered in fabric, 525w x 675d x 825h, with folding tablet arm.
S13	Examination table.

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S14	Medical bed with nurse and patient side rail control and patient lockout control

6.16.2.3 The description for credenzas, lateral files, storage cabinets, lockers, drawing storage cabinets, coffee tables and shelving is as follows:

ITEM NO.	DESCRIPTION
U1	Credenza, finely crafted wood, 1500w x 450d x 675h.
U2	Credenza, wood veneer, 1500w x 450d x 675h.
U3	Filing credenza, metal, 750w x 450d x 700h.
U4	Storage filing cabinet, metal, 750w x 450d x 1600h.
U5	Open-shelving filing cabinet, metal, 750w x 350d x 1600h.
U6	Filing cabinet for disks / cartridges, metal, 750w x 450d x 1600h.
U7	Vertical filing cabinet for drawings, metal, 840w x 935d x 1200h, top opening.
U8	Shelving cabinet with glass front, wood veneer, 750w x 450d x 1600h.
U9	Shelving cabinet with glass front, finely crafted wood, 750w x 450d x 1600h.
U10	Side table, wood, 600w x 600d x 450h.
U11	Coffee table, wood, 900w x 900d x 450h.
U12	Coffee table, wood, 550w x 1100d x 450h.
U13	Locker, type to COMPANY approval
U14	Highrise shelving (Refer Company Specification for Architectural Specialties, Metal, Wood and Plastics).
U15	Pallet racks (Refer Company Specification for Architectural Specialties, Metal, Wood and Plastics).
U16	Cantilever racks (Refer Company Specification for Architectural Specialties, Metal, Wood and Plastics).
U17	Storage cabinet, metal, 900w x 450d x 1980h, with locking doors.
U18	Built-in, laminated, adjustable shelving.
U19	Freestanding metal shelving with adjustable shelves.

6.16.3 Execution

6.16.3.1 General

a) CONTRACTOR shall provide and install furniture as per the approved list.

b) Upon completion of installation and adjustments, all labels shall be removed and all surfaces shall be cleaned as recommended by the manufacturer.

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c) All furniture shall be protected up to the time of inspection and acceptance by the COMPANY.

d) All cartons, packing materials and rubbish due to furniture work shall be removed.

6.16.3.2 Furniture, Process Buildings

Furniture to be provided as listed on Architectural Drawings.

6.16.3.3 Furniture, Non-Process Buildings

Furniture to be provided as listed on Architectural Drawings.

6.16.3.4 Refurbished Buildings

Furniture to be provided as listed on Architectural Drawings.

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SECTION C – OTHER REQUIREMENTS

7 QUALITY PLANS

7.1 General

a. Quality plans shall address all aspects related to local conditions, such as climatic conditions, backup facilities, spare parts, transport possibilities, storage facilities, local Manufacturers / Suppliers, and test facilities.

b. Quality Management Systems shall comply with the applicable requirements of BS EN ISO 9001 'Quality Management Systems - Requirements' and BS EN ISO 9004, 'Quality management – Quality of an organization – Guidance to achieve Sustained Success'. Written quality plans and procedures shall be submitted to COMPANY for review and approval.

c. Materials should be obtained from the same source(s) throughout the work.

7.2 Inspection, Testing and Reporting

a. Inspection and Test Plans (ITPs) identifying the tests, frequencies, acceptance criteria, and responsibilities shall be prepared as part of the quality plan.

b. A reporting system of Quality Control Records (QCRs) shall be part of the quality plan.

c. The reporting system shall record all results obtained during the testing.

d. Weather conditions shall be recorded in the quality records.

e. The proposed ITPs and QCRs shall be included in the Quality Plan.

f. Standard forms of inspection and test plans shall be used as a basis for the development of the ITPs and QCRs supplemented by the requirements of the Specifications for the work to be executed.

g. The COMPANY reserves the right to make inspections of the source of supply of materials.

h. Prior to supply of any material to site the CONTRACTOR shall obtain COMPANY approval by submitting necessary documentation such as test certificates, source of supply, date of manufacture, etc.

7.3 Quality Control and Assurance

a. CONTRACTOR shall be solely responsible for quality control of all materials and workmanship. The quality system shall provide for the planned and systematic control of all quality-related activities performed during design and construction of all components. Implementation of the quality system shall be in accordance with the Project Agreement, CONTRACTOR's approved Quality Manual and Project Specific Quality plan.

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b. CONTRACTOR shall ensure that SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERs have in place a QA / QC program which clearly establishes the authority and responsibilities of those responsible for the quality system. Persons performing quality functions shall have sufficient and well-defined authority to enforce quality requirements that they initiate or identify and to recommend and provide solutions for quality problems and thereafter verify the effectiveness of the corrective action.

c. Quality System and Quality Control requirements shall be identified and included in CONTRACTOR's Purchase documentation. Based on these requirements, the SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERS will develop QA / QC programs which shall be submitted to COMPANY and CONTRACTOR for review and concurrence. The SUB-CONTRACTOR's, SUPPLIER's and MANUFACTURER's QA/QC programs shall extend to SUB-SUPPLIERS.

d. CONTRACTOR shall submit certified reports of all tests as soon as the tests are completed satisfactorily.

e. COMPANY reserves the right to inspect or to conduct a quality audit of materials, installation and workmanship standards and shall have unrestricted entry to the shop of the fabricator at all stages of manufacture and to witness any or all tests. SUB-CONTRACTORs, SUPPLIERs, and MANUFACTURERs, within 30 days after award but prior to the pre-inspection meeting, shall provide CONTRACTOR and COMPANY with a copy of its Manufacturing and Inspection Plan for review and inclusion of any mandatory CONTRACTOR and COMPANY witness or hold points.

f. COMPANY may reject improper, inferior, defective, or unsuitable materials, installation, and workmanship. All materials and workmanship rejected shall be repaired or replaced by CONTRACTOR as directed by the COMPANY.

8 SUB-CONTRACTORS, SUB-SUPPLIERS

a. CONTRACTOR shall assume unit responsibility and overall guarantee.

b. CONTRACTOR shall transmit all relevant Purchase Order documents including COMPANY specifications to his SUB-CONTRACTORs and SUPPLIER's.

c. It is CONTRACTOR's responsibility to enforce all Purchase Order and Project Specification requirements on its SUB-CONTRACTORS and SUPPLIERS.

d. CONTRACTOR shall submit all relevant SUB-CONTRACTOR and SUPPLIER engineering data to COMPANY.

e. CONTRACTOR shall obtain and transmit all SUB-CONTRACTOR and SUPPLIER warranties to COMPANY.

f. All drawings and engineering data submitted to COMPANY shall be in the English language, and in metric units.

9 NOTIFICATION OF NONCONFORMANCE

The CONTRACTOR shall notify the COMPANY within 24 hours of any test, or any vendor supplied data that is required by this specification and does not meet the test limitations as shown COMPANY Standards listed in Section 2 of this document.

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10 DOCUMENTATION / MANUFACTURER DATA RECORDS

a. The CONTRACTOR and its SUBCONTRACTORs shall maintain test records and reports in accordance with the Standards as listed in Section 2.0.

b. The CONTRACTOR shall provide the COMPANY, within 24 hours, reports of all tests results.

c. The CONTRACTOR shall provide free access to COMPANY Representatives for supervision, inspection and qualitative check of the work and testing being executed.

d. In addition to test records and reports, the CONTRACTOR and its SUBCONTRACTORs shall review and maintain Manufacturer / Vendor data sheets, Material Certificates and Certificates of Conformance as required by the Standards as listed in Section 2.0.

11 HANDLING

11.1 Packaging and Shipping

a. Delivery of materials shall be in accordance with the MANUFACTURER / VENDOR'S instructions and as noted in the purchase order.

b. Preparation for shipment and packing will be subject to inspection and acceptance by the CONTRACTOR / COMPANY. All costs incurred by any rejection shall be at the expense of the MANUFACTURER / VENDOR.

c. The materials shall be shipped and delivered to the site in the VENDOR'S protective shipping containers. Should they need to be removed from their shipping containers, they shall be supported and covered so that they are protected from the elements and from adjacent construction activities.

d. Material shall be packed, securely anchored, and skid mounted when required.

e. Open ends of tubes and pipe shall be capped for protection. Separate, loose, and spare parts shall be completely boxed. A bill of materials shall be enclosed in each package or container of parts.

f. Exposed finished and machined surfaces, including bolting, shall be given a coating of rust inhibiting compound. Internal metal surfaces shall be sprayed or coated with a suitable rust preventative prior to shipment. Openings shall be suitably tagged to indicate the rust preventative applied.

g. Any finishes which become damaged shall be repaired in accordance with the respective Specification.

11.2 Preservation and Storage

a. Materials shall be protected to withstand an extended period of storage at the jobsite for a minimum period of 18 months. Materials shall be stored indoors and elevated, protected to safeguard against all adverse environments, such as humidity, moisture, rain, dust, dirt, sand, mud, salt air, salt spray and sea water.

b. All materials shall be preserved in accordance with Specifications for Preservation and Export Packing to COMPANY requirements and / or relevant codes and procedures.

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c. Items damaged due to improper handling and / or storage shall be replaced or repaired, at the CONTRACTOR'S option, by the SUBCONTRACTOR and at no cost to the COMPANY.

12 WARRANTY REQUIREMENTS FOR ARCHITECTURAL MATERIALS

Unless otherwise specified in Project documents, the following minimum warranty period shall be applicable for architectural materials.

a. Paint: Exterior and Interior - 5 years.

b. Glass and Glazing: Insulated glass units: Condensation, fogging and obstruction of vision due to deterioration of film on internal surfaces shall be considered as seals failure - 10 years. Coating on coloured glass: Coating shall be free of oxidation, de-lamination, colour change, streaks, disintegration and free of: Scratches, except those maximum 75 mm in length and visible at a distance of maximum 2000 mm. Pinholes, except those up to 1.2 mm in size. Large clusters of pinholes, as assessed by COMPANY are not permitted - 10 years. Reflective glass coating - 10 years.

c. Door Hardware: Hinges (including guarantee against wear) – 5 years. Locks – 5 years. Overhead surface mounted door closers – 5 years. Non-mechanical items like plates, handles etc: (subject to normal wear characteristics of the materials and finishes supplied according to these specifications) – 10 years.

d. Metal Door and Frames: Materials and workmanship – 10 years.

e. Wooden Doors: Defects in material and workmanship including warping, cupping, twisting, shrinkage, swelling, delamination, splitting and telegraphing of core – 10 years.

f. Aluminium Doors and Windows: Metal finish and glass – 10 years.

g. Membrane Roofing: Materials and workmanship. These shall be free of defects and shall comply with manufacturers technical specifications and instructions – 10 years.

h. Resilient Flooring and Skirting: Materials and Workmanship – 5 years.

i. Raised Access Flooring: Pedestals – 10 years. Floor panels – 10 years.

j. Cut Natural Stone: Materials and Workmanship – 10 years.

k. Carpet Tiles: Excessive surface wear – 10 years. Latent defect – agree with COMPANY. Pattern match – agree with COMPANY. Floor compatibility – 10 years. Anti-static – 10 years.

l. Fluid applied Waterproofing: Materials and Workmanship – 10 years.

m. Joint Sealants: Materials and Workmanship. Defects shall include but shall not be limited to drying and failure in adhesion or cohesion, air and moisture leakage, discolouration and staining of adjacent materials – 5 years.

n. Thermal Protection: Materials and Workmanship – 5 years.

o. Fire Stopping: Materials and Workmanship – 5 years.

p. Termites: If subterranean termite activity or damage is discovered during warranty period, re-treat soil and repair or replace damage caused by termite infestation – 5 years.

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SECTION D – STANDARD DRAWINGS & DATASHEETS

13 DATASHEET TEMPLATES

Not applicable.

14 STANDARD DRAWINGS

Not applicable

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SECTION E – APPENDICES

Not applicable.

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